



UNIVERSITÀ DI PISA

Data Mining 2 Project Report

Exploring the CMI Dataset

A Study on Data Understanding, Outlier Detection, Classification, Regression and Explainability

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1. Introduction

This project analyzes the Child Mind Institute dataset to identify predictors of problematic internet use in children and adolescents, focusing on the Severity Impairment Index (SII). To ensure reliable, interpretable results, we implement a structured data mining pipeline: **Module 0** prepares the data through quality assessment, imputation, and feature engineering; **Module 1** addresses core data challenges, including outlier detection and class imbalance; and **Module 2** applies supervised learning algorithms to predict the SII target, evaluating model performance and behavior.

2. Module 0: Data Understanding and Preparation for Tabular Dataset

2.1 Dataset Overview

This multimodal dataset tracks 8,460 participants aged 5 to 22 across 82 attributes, including physical measurements, body composition (BIA), fitness (FGC), sleep (SDS), and physical activity (PAQ). These features combine physiological and behavioral data to predict the Severity Impairment Index (sii), an ordinal clinical target ranging from 0 (no impairment) to 3 (severe impairment).

2.2 Target Variable Distribution

The Severity Impairment Index (sii) target is highly imbalanced, with most observations concentrated in the class (sii = 0) and few in the severe impairment class (sii = 3). This severe skew makes identifying minority cases difficult, requiring specialized class imbalance techniques in subsequent analysis steps.

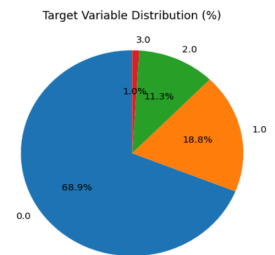


Fig 1: Distribution of the target variable (SII)

2.3 Data cleaning

2.3.1 Impossible values

Before handling missing values, we identified and replaced biologically or clinically impossible values with NaN. Data points were only flagged as invalid if they strictly violated medical constraints, ensuring genuine statistical extremes were preserved.

The following impossible values were detected and corrected:

Table 1: Summary of impossible values identified based on biological and clinical constraints

Feature	Impossible condition	Count	Action
CGAS-CGAS_Score	0 < or > 100	16	NaN
Physical-BMI	== 0	7	NaN
Physical-Weight	== 0 lbs	81	NaN
Physical-Diastolic_BP	== 0 mmHg	1	NaN
Physical-Systolic_BP	== 0 mmHg	1	NaN
Physical-Height	< 36 inches	1	NaN
Physical-HeartRate	< 30 bpm	1	NaN
BIA-BIA_Fat	< 0% or > 60%	389	NaN
BIA-BIA_FMI	< 0 or > 30	61	NaN
BIA-BIA_FFMI	< 5 or > 30	38	NaN
BIA-BIA_SMM	< 5 or > 100 kg	65	NaN
BIA-BIA_FFM	< 10 or > 150 kg	83	Nan

These entries were considered **invalid** because they violate fundamental physiological limits. Measurements like zero body weight, zero blood pressure, or negative body fat are biologically

impossible, while other extreme values fall entirely outside realistic human ranges. Likely caused by recording or **measurement errors**, these entries were treated as missing values.

2.3.2 Dropping Columns

All **22 PCIAT** variables were excluded due to data leakage (since they highly correlate with the target ($r = 0.90$) and would cause circular prediction and severe missingness (67.9%), where large-scale imputation would introduce artificial noise. Highly correlated features ($|r| > 0.75$) were eliminated to address multicollinearity. For example, **BIA-BIA_BMI** was dropped in favor of the standard Physical-BMI, and **SDS_Total_Raw** was replaced by **SDS_Total_T**, **multiple collinear BIA metrics** (e.g., BMC, BMR, TBW) were similarly removed due to overlap with retained lean mass and fat variables. **Lastly**, variables with negligible correlation to the target (such as **FGC_TL**, **Endurance_Time_Total**, and **PAQ_Total**) and **redundant categorical** representations (**FGC Zone** and **Season** features) were excluded to finalize a compact, predictive feature set.

2.4 Handling Missing and Duplicate Values

There was no duplicates values found in dataset. Missing data showed a **block-wise pattern**: when a participant missed a session, all variables related to that screening were missing. The highest missingness rates were found in **SDS**, **Fitness Endurance** (35.2%), **PAQ** (24.5%–38.2%), and **FGC** (17.9%–31.7%), while **CGAS**, **internet usage**, **Physical**, and **BIA** variables had lower missingness. This suggests a **Missing Not At Random (MNAR)** pattern, where data are missing because the screening itself was not completed. Since applying the same imputation method to all variables could introduce bias, different strategies were used for each block to preserve the data structure and reduce distortion.

2.4.1 KNN Imputation

For physical and biometric parameters, a block-wise K-Nearest Neighbors (KNN) algorithm (**k=5**) was applied to reconstruct realistic values based on subject similarity.

- **Physical Block (9.7%–33.5% missing)**: Height, weight, BMI, blood pressure, waist circumference, and heart rate were estimated using the values of the 5 most physically similar subjects.
- **BIA Block (21% missing)**: Corrupted and biologically impossible values were removed, and remaining data was capped between the 1st and 99th percentiles. KNN then reconstructed lean and fat mass values based on the participant's general physical structure, preserving underlying physiological relationships.

2.4.2 Median Imputation

For blocks with higher missing rates, behavioral data, or weaker feature dependencies, the overall median was applied as a robust estimate to avoid model bias.

- **SDS (42% missing)**: The median was used to fill missing values because it is robust and does not strongly affect the original data distribution.
- **PAQ (24.5%–38.2% missing)**: Missingness reflects separate rates for **PAQ_A** and **PAQ_C** before they were unified into the single **PAQ_Total** feature, then the residual missing values were filled with the median. **PAQ_Total** was later removed due to low predictive value and redundancy
- **FGC (17.9%–31.7%), CGAS (16.9%), and Computer/Internet Hours (7% missing)**: were filled using the median because these variables showed moderate levels of missingness and had limited relationships with other features. This provided a simple and reliable way to complete the dataset
- **Fitness Endurance Max Stage (35.22% missing)**: because physical performance is highly unpredictable from other features, the median served as a conservative baseline to avoid overestimating participant capabilities.

2.4.3 Missingness indicator

To account for non-random missingness, two binary indicators (**BIA_measured** and **SDS_measured**) were retained to track whether these blocks were originally observed or imputed.

This preserves the original missingness pattern, helping the model distinguish between actual and synthetic data. Ultimately, this captures systematic participant differences, increasing model robustness and minimizing imputation bias.

2.5 Feature Engineering

In this stage, new features were constructed to reduce redundancy, improve interpretability, and capture more informative representations of the underlying physiological and behavioural patterns. Symmetric and complementary measurements were aggregated to reduce noise while preserving relevant signal.

- **FGC_Flexibility_Avg**: Average of left and right sit-and-reach tests (**FGC_SRL**, **FGC_SRR**), which measure the same construct ($r = 0.70$); averaging reduces noise while preserving flexibility.
- **FGC_Grip_Avg**: Average of dominant and non-dominant hand grip strength (**FGC_GSD**, **FGC_GSND**), providing a more stable estimate of upper body strength ($r = 0.65$).
- **FGC_Muscular_Endurance**: Average of curl-ups and push-ups (**FGC_CU**, **FGC_PU**), capturing overall muscular endurance; this feature shows meaningful correlation with *sii* ($r = 0.187$).
- **PAQ_Total**: Combined physical activity score from **PAQ_A** and **PAQ_C** using a coalesce operation, unifying measurements across age groups while preserving all available data, which was dropped later.

2.5.1 Derived Features

Additional composite features were constructed to capture higher-level behavioral and physiological relationships that are not directly observable from individual variables.

1. Social_Digital_Displacement = $\text{Internet_Hours} / (\text{CGAS} / 10 + 1)$. Measures the degree to which internet use displaces normal social functioning. Higher screen time combined with lower psychosocial functioning (CGAS) produces higher values. The +1 in the denominator prevents division by zero.

2. Lifestyle_Risk_Score = $(\text{SDS_Total_T} / 100) + 1 / (\text{PAQ_Total} + 1) + (\text{Internet_Hours} / 10)$. A composite behavioural risk index combining emotional distress (SDS), physical inactivity (PAQ), and screen time. Each component is scaled to a comparable range before summation.

3. Metabolic_Alarms_Count: An integer score from 0 to 5 representing the number of metabolic risk flags triggered for each child. The count increases by one for each of the following conditions met: $\text{BMI} > 25$, $\text{Systolic_BP} > 120$, $\text{PAQ_Total} < 2$, $\text{Body_Fat} > 30\%$, and $\text{Fitness_Endurance_Stage} < 3$. The final score is the sum of all triggered flags.

4. ScreenTime_Age_Ratio = $\text{Internet_Hours} / \text{Age}$. Contextualises screen time relative to developmental stage, assigning higher risk to younger individuals with similar usage levels.

4. ScreenTime_Category: Ordinal grouping of daily internet hours into four risk tiers as 0 (low: < 2 hours), 1 (moderate: 2–4 hours), 2 (high: 4–6 hours), 3 (very high: ≥ 6 hours).

2.6 Preprocessing Summary

After preprocessing, cleaning, imputation, and feature engineering, the final dataset contains **8,460 observations and 32 attributes**. The data is complete, compact, and suitable for analysis, with all features represented in numeric or ordinal form and redundant variables removed to reduce noise. The original distribution of the **SII** target variable is preserved. The processed dataset is then used for **outlier and class imbalance analysis**, followed by **classification and regression modelling**.

3. Module 1: Outliers

To identify anomalous observations, three complementary outlier detection methods were applied to capture deviation-based, density-based, and distance-based anomalies: **Histogram-Based Outlier Score (HBOS)**, **Local Outlier Factor (LOF)**, and **K-Nearest Neighbors (KNN)**.

Prior to detection, several preprocessing choices were made to protect the integrity of the algorithms. The target variable (*sii*) was excluded to prevent data leakage, while binary indicators,

ordinal columns, and categorical features (such as BIA_measured, SDS_measured, and ScreenTimeCategory) were removed because they distort distance and density calculations. The remaining **28 numeric features** were then standardized using Z-score normalization (**StandardScaler**) to ensure equal contribution across different scales, providing a consistent matrix for all three methods.

A contamination rate of **1%** was set for each algorithm. To maximize robustness, a consensus-based ensemble approach was adopted: an observation was classified as an outlier and removed only if flagged by **at least two of the three methods**. Finally, the cleaned dataset was saved in its original, unstandardized scale to preserve interpretability for downstream analysis.

3.1 Histogram-Based Outlier Score (HBOS): a univariate deviation-based method

The Histogram-Based Outlier Score (HBOS) algorithm assigns anomaly scores based on the density of values in each feature. Observations located in low density (rare) regions across multiple variables receive higher scores and are more likely to be detected as outliers. Because HBOS assumes strict feature independence, a Principal Component Analysis (PCA) step was applied first. This transformed strongly correlated variables (height, weight, BMI, and BIA data) into orthogonal components, preventing redundant features from artificially inflating the anomaly scores.

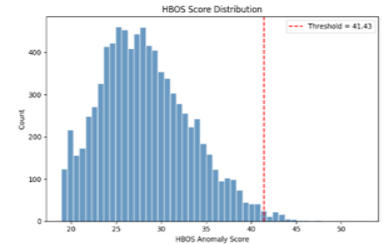


Fig 2: Distribution of HBOS anomaly scores

By retaining components that account for 95% of the explained variance, the model preserved original information while establishing independence. With a 1% contamination threshold, HBOS identified 85 outliers. The **10 highest-scoring HBOS outliers** (scores ~49-56, all flagged label=1), with row index 59 being the most extreme, well above the closest score. Moreover, HBOS-flagged outliers differ from normal children most strongly in metabolic alarm count, muscular endurance, fat mass index, weight, and grip strength, all sharply elevated, while only four features (flexibility, CGAS score, sex, and activity level) are slightly lower than typical children, suggesting these outliers are big, strong, but marginally less active/functional kids rather than underweight ones.

3.2 Local Outlier Factor (LOF): a density-based method

Next, the Local Outlier Factor (LOF) algorithm was applied. As a density-based method, LOF identifies anomalies by comparing the local density of an observation against its neighbors, allowing it to detect points that are isolated relative to their specific surrounding region rather than the global dataset.

For each observation, LOF computes the local reachability density (LRD), which measures how densely a point is clustered relative to its k-nearest neighbors; points in significantly lower-density regions receive higher LOF scores. The neighborhood parameter k was selected via sensitivity analysis across **k in {10, 20, 50}**. A value of **k=10** was unstable due to sensitivity to near-duplicate observations, while **k=50** overly smoothed local density structures. Setting **k=20** provided the optimal balance, preserving local variations while maintaining algorithmic stability. With a 1% contamination rate, LOF identified 85 outliers based on the threshold separating the top 1% of anomaly scores.

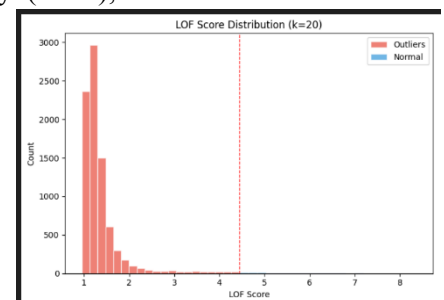


Fig 3: Distribution of LOF anomaly scores

3.3 K-Nearest Neighbors (KNN): a distance-based method

Finally, the k-Nearest Neighbors (KNN) outlier detection method was applied. As a distance-based approach, KNN identifies anomalies by calculating the distance from an observation to its k-th nearest neighbor, detecting global outliers that are far from the majority of the data distribution. The neighborhood parameter was set to **k=30** following an elbow analysis over the range **k in**

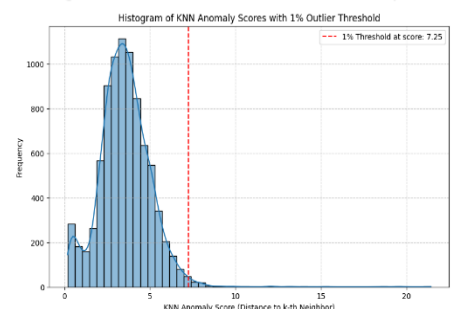


Fig 4: Distribution of KNN anomaly scores

{5, 10, 20, 30, 50} Inspecting the sorted k-distance curves revealed a clear, stable inflection point at k=30 that cleanly separated dense normal clusters from isolated data points. Applying a 1% contamination rate, the algorithm flagged the top 1% of anomaly scores, successfully identifying 85 outliers.

The analysis shows that **outliers differ** significantly from the rest of the population across several key characteristics. Outliers generally exhibit higher values for body weight (Physical-Weight), fat mass index (BIA-BIA_FMI), lean mass (BIA-BIA_FFM), skeletal muscle mass (BIA-BIA_SMM), and metabolic risk indicators (Metabolic_Alarms_Count). Elevated waist circumference, blood pressure, age, and internet usage are also observed. Overall, this pattern suggests a profile associated with overweight/obesity, increased metabolic risk, and, in some cases, greater use of digital devices.

3.4 Comparison of Detection Methods and Visual Interpretation

To assess agreement among the three outlier detection methods, we analyzed their overlapping detections and their spatial distribution using a common PCA projection.

HBOS and **KNN** agreed on **42** severely obese adolescents (mean BMI 34.1, waist 34" vs normal 27", internet 2.2 hrs/day vs 1.0, sleep disturbance 67.5 vs 56.7)exceeding multiple clinical thresholds. **LOF uniquely flagged 85** young lean child (mean weight 69.5 lbs vs normal 84, body fat 15% vs 19%, muscle mass 27.4 vs 30.7, all aged 5–10)who are developmentally normal but locally isolated. Zero consensus outliers across all three methods, no single case exceeded all detection thresholds simultaneously, proving outlier status is algorithm-dependent.

As shown in **Figure 5 and Table 1.2**, there is limited agreement between the methods. Although each algorithm identified 85 outliers based on the 1% contamination threshold, only **42 observations** were jointly detected by both **HBOS and KNN**, while LOF detections showed no overlap with either. This lack of convergence indicates that each method captures distinct types of anomalous behavior rather than a single set of outliers.

On the **PCA visualization**, the consensus outliers flagged by **at least two methods** appear as isolated points in the feature space, confirming they represent genuine structural deviations rather than dense clusters or systematic noise.

The **LOF method** identified some observations from younger participants as outliers because they showed unusual combinations of physical and behavioral characteristics. However, these differences may simply reflect normal developmental variation rather than true anomalies. This may explain why LOF produced different results from the other methods. It also shows a limitation of density-based approaches: in datasets with missing values and imputed data, local density estimates can become less reliable.

In contrast, global methods like **KNN** and **HBOS** provided more stable detections because they rely on overall distance or feature-wise deviations.

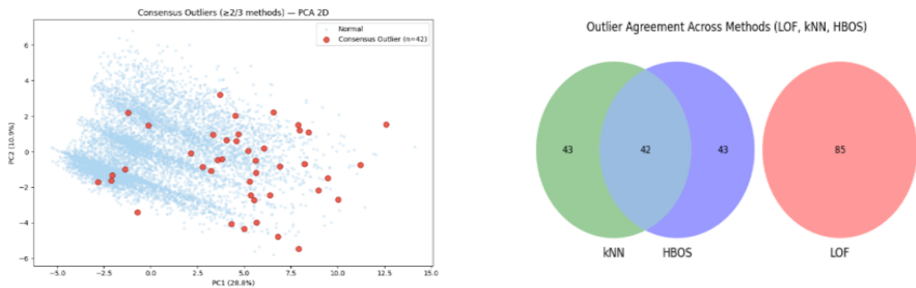


Figure 5: PCA projection of the dataset highlighting consensus outliers (flagged by at least two methods), showing their scattered and isolated distribution in the feature space.

Table 1.2. Comparison of the three outlier detection methods and their consensus overlap.

Method	Type	Key Parameter	Outliers (1%)	Consensus Overlap
HBOS on PCA	Statistical (histogram)	PCA 95% var. Cont. = 1%	[85]	[42] (with KNN)
LOF	Density (local)	k = 20, cont. = 1%	[85]	None
KNN	Distance (global)	k = 30, 99th pct	[85]	[42] (with HBOS)

3.5 Consensus Outlier Removal

After detecting the top 1% anomalies using HBOS, LOF, and KNN, a majority-vote consensus strategy was applied. An observation was confirmed as an outlier if it was flagged by at least two of the three methods. This approach reduces false positives, as observations detected by only a single method are more likely to represent method-specific artefacts rather than genuine anomalies.

In total, **42 consensus outliers** were identified and removed from the dataset. In addition, a manual verification step was performed for physically extreme observations. Records with **Physical_Weight $\geq$ 250 lbs** were cross-checked by recomputing BMI from height and weight. One case, corresponding to a 7-year-old child with a recorded weight of 256 lbs, exhibited a significant inconsistency between the recorded and recomputed BMI, indicating a **data entry error**. This record was manually removed, as it was not captured by the automated consensus process.

The PCA visualization in **Figure 6** confirms the validity of the removal process. The detected outliers are located at the periphery of the data distribution, indicating that they represent genuinely anomalous observations rather than noise within dense regions.

The resulting cleaned dataset contains **8,417 observations and 31 features**, providing a compact and reliable foundation for subsequent analysis.

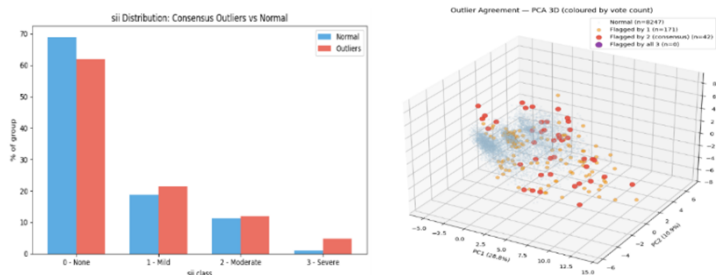


Figure 6: 3D PCA projection after consensus-based outlier removal. Confirmed outliers appear at the periphery of the data cloud, validating that the removed observations represent genuine anomalies.

4 Module 1: Imbalanced Learning

To mitigate bias from the highly imbalanced *sii* target (8,417 samples: 69% class 0; 1% class 3), we applied an 80%-20% stratified split (6,733 train, 1,684 test). We evaluated Decision Tree and KNN classifiers, optimizing hyperparameters via GridSearchCV (5-fold, weighted F1). KNN inputs were Z-standardized on the training set. To improve minority class detection, we tested RUS, ROS, Tomek Links, and SMOTE-NC exclusively on the training set.

4.1 Baseline Classification without Imbalance Techniques

We evaluated the Decision Tree and KNN models on the original imbalanced dataset using GridSearchCV. Optimal hyperparameters identified were: Decision Tree (criterion=gini, max_depth=5, min_samples_split=2) and KNN (metric=manhattan, n_neighbors=5, weights=distance). **Table2** summarizes the performance, illustrating the impact of class imbalance: models performed well on the majority 'None' class but struggled with low recall for minority classes, particularly 'Severe'.

Table 2: Baseline performance on imbalanced dataset (Weighted Averages)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision Tree	0.6811	0.5763	0.6811	0.5987	0.6680
KNN	0.6568	0.5774	0.6568	0.5986	0.6167

4.2 Balancing Techniques for Imbalanced Learning

We evaluated the effectiveness of the five balancing strategies on the Decision Tree and KNN classifiers. Table 3 presents the comparative performance metrics for these methods.

Table 3: Performance comparison after applying imbalance handling techniques

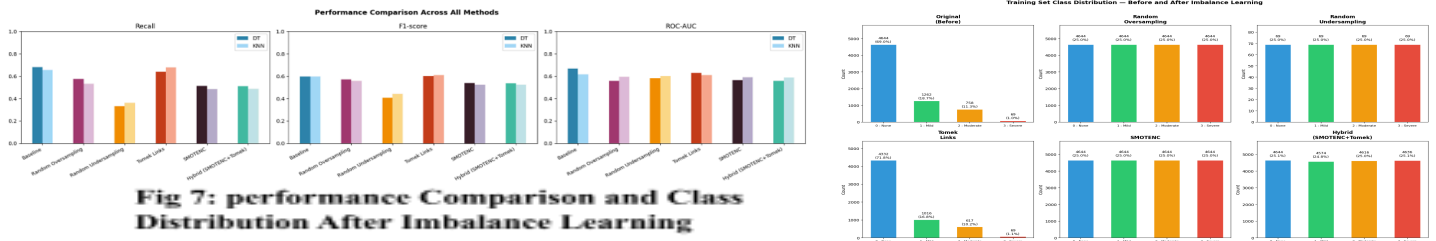
Method	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Oversampling	Decision Tree	0.5760	0.5698	0.5760	0.5728	0.5597
	KNN	0.5321	0.5962	0.5321	0.5578	0.5941
Random Undersampling	Decision Tree	0.3325	0.5906	0.3325	0.4064	0.5826
	KNN	0.3622	0.6096	0.3622	0.4428	0.6009
Tomek Links	Decision Tree	0.6390	0.5768	0.6390	0.6016	0.6297
	KNN	0.6781	0.5945	0.6781	0.6105	0.6117
SMOTENC	Decision Tree	0.5143	0.5758	0.5143	0.5400	0.5633
	KNN	0.4852	0.5810	0.4852	0.5226	0.5899
Hybrid (SMOTENC+Tomek)	Decision Tree	0.5089	0.5718	0.5089	0.5354	0.5587
	KNN	0.4869	0.5823	0.4869	0.5243	0.5890

4.3 Discussion and Impact Analysis

- **Random Undersampling (RUS):** Reduced the training set from 6,733 to 276 rows (matching the 69 Severe samples). Test accuracy dropped sharply to 33%–36% due to data loss, but achieved the highest Severe recall of all methods (DT: 47%, KNN: 53%). This is the only strategy that effectively prioritized severe case detection.
- **Tomek Links:** Removed 699 borderline majority samples, reducing the training set to 6,034 rows. It yielded the highest test F1-score (KNN: 0.6105, DT: 0.6016) by cleaning the decision boundary, though Severe recall remained low at 0%–6%.
- **Random Oversampling (ROS):** Replicated minority instances to balance all classes at 4,644 samples each (18,576 total rows). It overfitted to the training data, lowering test performance below baseline (DT F1: 0.57, KNN F1: 0.56) and yielding a Severe recall of only 6%.
- **SMOTE-NC:** Interpolated ~12,000 synthetic samples to balance all classes at 4,644 instances. Test accuracy dropped to 51% (DT) and 49% (KNN) as synthetic patterns failed to generalize well, though Moderate recall improved to 26%–29% and Severe recall reached 6%.
- **Hybrid (SMOTE-NC+Tomek):** Post-SMOTE Tomek filtering removed just 106 boundary samples (0.6% of the 18,576 rows). Metrics remained nearly identical to SMOTE-NC alone (DT F1: 0.54, KNN F1: 0.52), indicating few synthetic points fell in ambiguous zones.
- **Confusion Matrix Analysis:** While oversampling methods (ROS, SMOTE-NC) increased minority detection, they severely misclassified the majority "None" class compared to boundary-cleaning approaches like Tomek Links.

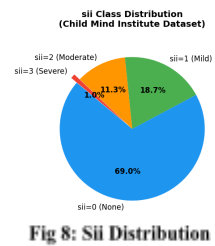
The **original training set** is severely imbalanced (69% None vs only 1% Severe), so ROS and SMOTENC fully balance all four classes to 25% each by generating new minority samples, while RUS achieves the same balance by aggressively shrinking the majority class down to just 69 samples. Tomek Links barely changes the distribution since it only removes borderline noise rather than balancing classes, and the Hybrid approach (SMOTENC + Tomek) ends up nearly

balanced (~25% each) since Tomek's cleanup after oversampling removes only a small number of ambiguous synthetic points.



5 Module 2: Advance classification

This module applies advanced classification algorithms to the cleaned dataset (8,417 samples, 32 features) to predict the ordinal sii target (0–3). To address the severe class imbalance (Class 0: 69%, Class 3: 1%), we performed an **80%–20%** stratified split and optimized hyperparameters using **5-fold Stratified Cross-Validation** scored on macro F1. Preprocessing was model-specific: Z-score scaling was applied to distance/gradient-sensitive classifiers (Logistic Regression, SVM, Neural Networks), while scale-invariant ensemble methods (Random Forest, XGBoost, AdaBoost) utilized raw features. **Table 4** details the hyperparameter search spaces.



5.1 Class Grouping Strategy: From 4-Class to 3-Class Problem

The extreme scarcity of the Severe class (sii=3, ~1% of data) caused all 4-class models to achieve 0% recall on severe cases, as the 69 training instances provided insufficient signal against the overlapping Moderate class. To address this, it was reformulated the problem into a 3-class configuration by merging Moderate (sii=2) and Severe (sii=3) into a single **High Risk** class (sii=2). The resulting distribution is: None (69.0%), Mild (18.7%), and High Risk (12.3%). The documented results for both configurations to evaluate this structural adjustment are in **Table 5**.

Table 4: Hyperparameter Grids Used for Each Model

Model	Search Type	Hyperparameters
Logistic Regression	Grid Search	param_grid=[C:(0.01,0.1,1,10,100),penalty: (l2),solver:(lbfgs),(multi_class: (ovr, multinomial)),class_weight: (balanced,)] max_iter:(1000)]
SVM	Grid Search	param_grid=[(kernel:linear,rbf),(C:0.1,1,10,100),(gamma: scale,0.01, 0.1),(class_weight: ('balanced'))]
Random Forest	Grid Search	n_estimators: {50, 100, 200, 300}, max_depth: {5, 10, 20}, min_samples_split: {3, 7, 10}, min_samples_leaf: {1, 4, 6}, class_weight: {balanced}
XGBoost	Grid Search	n_estimators: {100, 200}, max_depth: {3, 5, 7}, learning_rate: {0.05, 0.1}, subsample: {0.8, 1.0}, colsample_bytree: {0.8, 1.0}
AdaBoost	Grid Search	n_estimators: {50, 100, 200}, learning_rate: {0.01, 0.1, 1.0}, estimator: DT(max_depth ∈ {1, 2, 3})
Neural Network	Randomized Search	Hidden layer sizes: {(256,128,64), (512,128,64), (128,64,32)}, activation: {relu, tanh}, solver: {adam, sgd}, alpha: {0.0001, 0.001, 0.01, 0.05}, learning_rate_init: {0.0001, 0.001, 0.01}, batch_size: {32, 64, 128}, learning_rate: adaptive

5.2 Logistic Regression

Logistic Regression was tuned using $C \in \{0.01, 0.1, 1, 10, 100\}$ with l2 penalty, lbfgs solver, and class_weight='balanced', testing both ovr and multinomial strategies. Despite this, the model

struggled with minority classes (sii=2 and 3), collapsing predictions into majority classes due to linear decision boundary limitations. As shown in **Figures 9-10**, this results in poor recall and lower ROC-AUC compared to non-linear alternatives. Summary metrics are provided in **Table 5**.

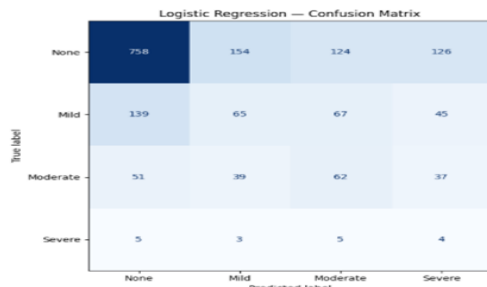


Figure 9: Confusion matrix Logistic Regression

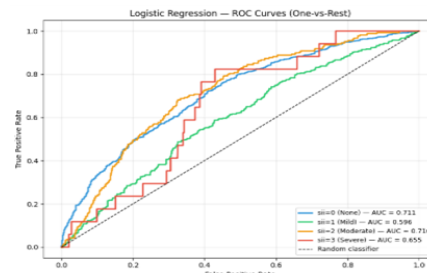


Figure 10: ROC curve Logistic Regression

5.3 Neural Network

To establish a **4-class baseline**, RandomizedSearchCV was performed on the original target variable (sii 0–3). Tuning was done using hidden layer sizes, activation functions (relu, tanh), solvers (adam, sgd), and learning rates. Training stability was ensured using L2 regularization (alpha=0.05) and an adaptive learning rate, these choices successfully controlled overfitting confirmed by the learning curves (**Figure 12**) showing steady loss decrease and stable validation score with no significant divergence. Despite this stability, the model yielded poor precision and recall for the minority classes (sii 1–3) due to the dominant majority class.

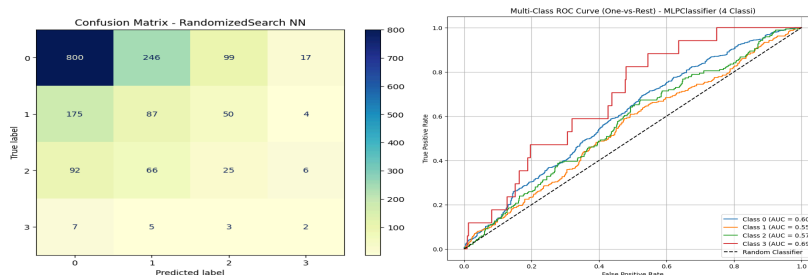


Fig 11: Confusion Matrix and ROCcurve Neural Network

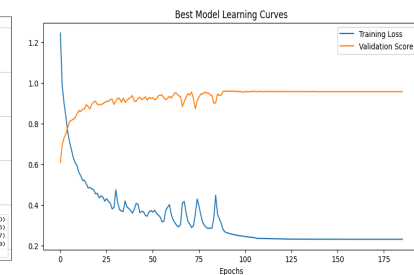


Fig 12: Learning curve

RandomizedSearch was performed using the same parameter ranges for the 3-class model to isolate the effect of class merging. This configuration proved robust within our explored ranges (**Figure 13-14**). While minority classes remain challenging, merging them improved category distinction.

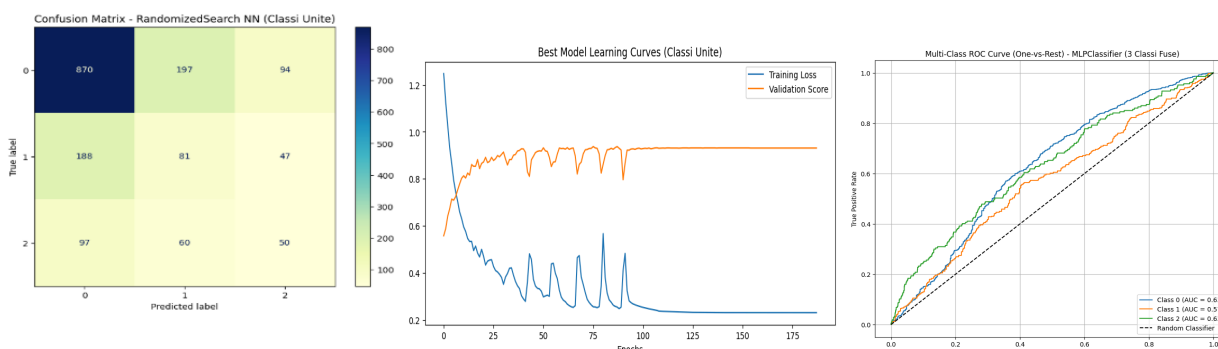


Fig 13: Confusion Matrix

Fig 14: ROCcurve Neural Network (Merged) and Learning curve (merged)

5.4 Random Forest

Random Forest model was optimized using **GridSearchCV**, tuning `n_estimators` (50–300) and `max_depth` (up to 20) to balance complexity and generalizability. Constraints on

min_samples_split and min_samples_leaf were strictly applied to prevent overfitting. To mitigate class imbalance, class_weight='balanced' was used **and** optimized for f1_macro. In the **4-class configuration**, the model performed well on the majority class (0) but struggled with rarer classes.

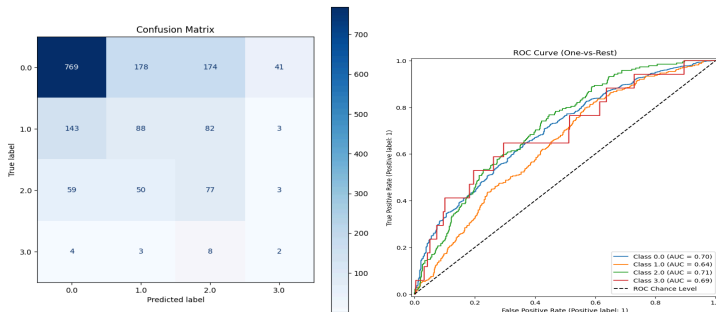


Figure 15: Confusion Matrix and ROCcurve Random Forest

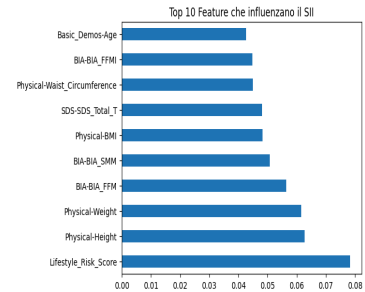


Figure 16: Top features analysis

To improve class structure, class 2 and 3 were merged into a single category, resulting in a **3-class model**. While the model achieved good precision and recall for class 0, performance on the less represented classes (1 and 2) remained limited, despite the use of class_weight = 'balanced' to mitigate class imbalance.

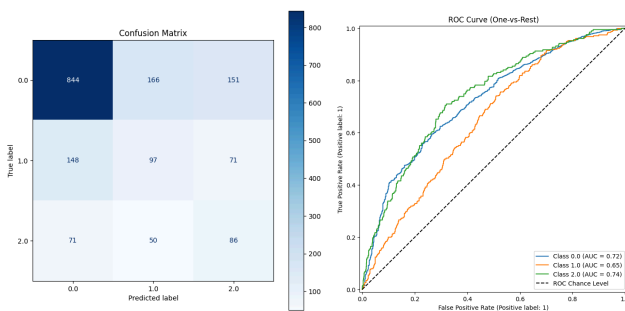


Figure 17: Confusion Matrix and ROCcurve Random Forest (merged)

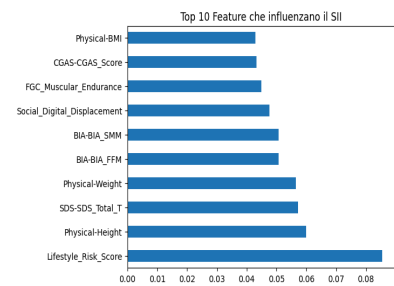


Figure 18: Top features analysis

The **Lifestyle Risk Score** is the primary predictive driver for sii. Higher behavioral risk scores directly correlate with increased classification probability for severe categories (sii=2 or 3), reinforcing the need for targeted lifestyle interventions.

5.5 SVM

For the Support Vector Machine (SVM), we evaluated both linear and RBF kernels, tuning $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{'scale', 0.01, 0.1\}$ to optimize the margin error trade-off and boundary smoothness. To address class imbalance, **class_weight='balanced'** was fixed across all runs.

In the **4-class configuration**, the RBF kernel better captured non-linear data structures, resulting in improved minority class performance (**Figure 19**). However, recall remained limited due to the severe imbalance.

To improve discriminative power, a **3-class formulation** was tested, merging Moderate (sii=2) and Severe (sii=3) into a single At-Risk class. This clinically justified step reduced the imbalance ratio, resulting in improved recall for the **At-Risk** category compared to the 4-class model (**Figure 20**). Summary metrics for both configurations are provided in **Table 5**.

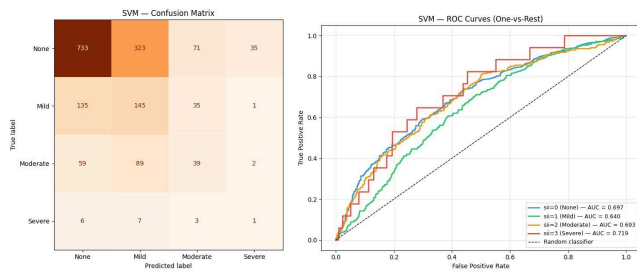


Fig 19: ConfusionMatrix and ROCcurve SVM

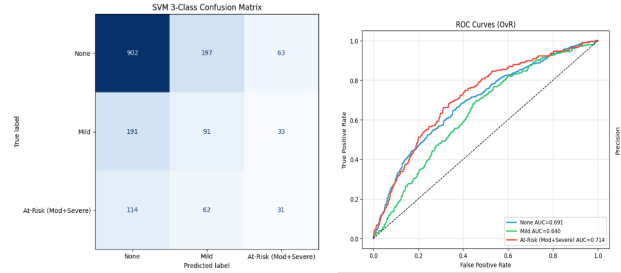


Fig 20: ConfusionMatrix and ROCcurve SVM (merged)

5.6 XGBoost

XGBoost employs gradient boosting to sequentially minimize residuals. We performed GridSearchCV over `n_estimators` (100–200), `max_depth` (3,5,7), `learning_rate` (0.05–0.1), and subsampling (0.8,1.0). To address imbalance, we use `class_weight=balance` to upweight minority classes without altering the dataset.

In the **4-class** setup, the model achieved 0.73 recall on the majority class but only 0.60 on the severe class (**Figure 21**). Merging Moderate and Severe into a **3-class "High Risk"** configuration significantly improved discriminative power. Optimal parameters shifted to `colsample_bytree=0.8`, `learning_rate=0.05`, `max_depth=7`, and `n_estimators=100`. The deeper trees and lower learning rate better captured the High Risk class boundary. This intervention improved test F1 Macro to 0.4436 (+0.105 vs 4-class) and correctly identified 29 of 207 High Risk cases (**Figure 22**). Summary metrics are provided in **Table 5**.

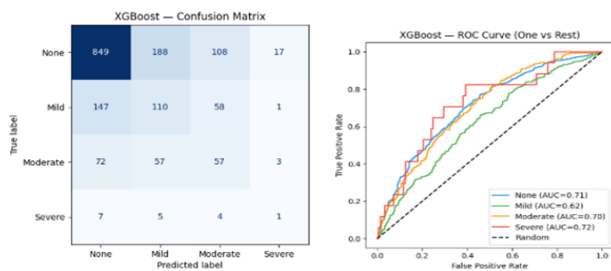


Figure 21: ConfusionMatrix and ROCcurve

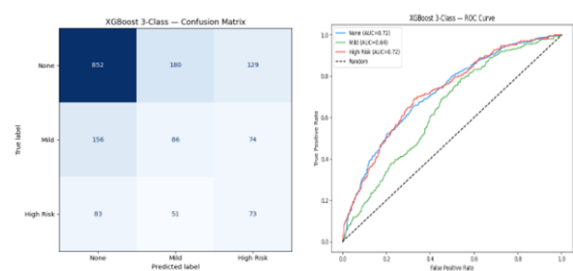


Figure 22: ConfusionMatrix and ROCcurve XGBoost (merged)

5.7 AdaBoost

AdaBoost (Adaptive Boosting) sequentially trains shallow decision trees, iteratively increasing the weight of misclassified samples. We utilized the SAMME algorithm with `class_weight='balanced'`, optimizing `n_estimators` (50,100,200), `learning_rate` (0.01,0.1,1.0), and base `max_depth` (1,2,3). The **4-class** configuration (estimator=DT(depth=3), lr=1.0, n=200) achieved an F1 Macro of 0.3023. The **3-class** AdaBoost model achieved its strongest performance on the None class, while substantial confusion was observed between the Mild and Moderate classes, resulting in weaker discrimination among impaired subjects. Notably, **Figure 23** reveals how AdaBoost's internal up-weighting mechanism differs from XGBoost in managing minority class errors. Summary metrics are provided in Table 5.

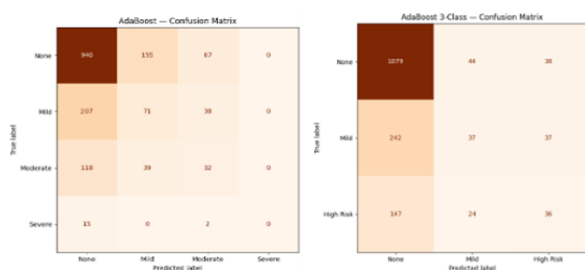


Figure 23: ConfusionMatrix and ROC(AdaBoost)

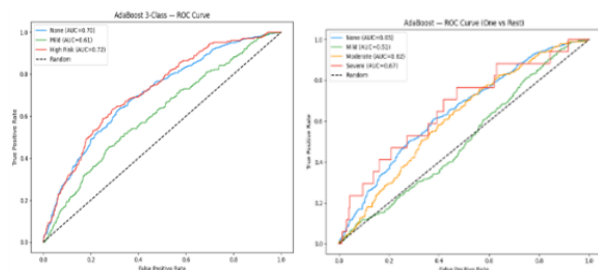


Figure 24: ConfusionMatrix and ROC(AdaBoost merged)

5.8 Conclusion

Table 5 synthesizes the hyperparameter and performance metrics for all models. **Random Forest** emerged as the top performer, achieving a maximum macro F1-score of 0.47 in the 3-class configuration, closely followed by XGBoost (0.4436) and AdaBoost (0.4077).

The transition from the **4-class to 3-class** formulation was the most critical analytical pivot of Module 2. In the initial 4-class configuration, all models exhibited **near-zero recall for the 'Severe' class**, with the sole exception of Logistic Regression, which only achieved detection through aggressive but imprecise class weighting. By merging 'Moderate' and 'Severe' into a 'High Risk' class, it achieved a systemic improvement in F1-score (+0.10 to +0.15 across all models) and unlocked meaningful recall for minority cases.

Regarding model performance:

- **Ensemble methods** (Random Forest, XGBoost) outperformed simpler architectures, benefiting from decorrelated trees and class-balancing logic.
- **AdaBoost** distinguished itself by achieving the best 'High Risk' recall (17%) through its iterative focus on difficult-to-classify samples.
- **Non-linear classifiers** (SVM with RBF kernel) consistently outperformed linear counterparts, proving that decision boundaries in this dataset are inherently non-linear.

Ultimately, these results confirm that the primary bottleneck for this dataset is not algorithmic, but rather data scarcity in the minority classes. The success of ensemble methods on the **3-class 'High Risk'** target demonstrates that while physical and behavioral features provide a foundational signal, clinically-informed class grouping is essential to achieve predictive utility.

Table 5: Best Hyperparameters and Performance Metrics for Each Model

Model	Best Params	Acc.	Macro F1	Macro Prec.	Macro Rec.	ROC-AUC
Logistic Regression (4-class)	C=0.01, l2, lbfgs, OvR, balanced	0.5279	0.31	0.33	0.36	0.6897
SVM (4-class)	rbf, C=1, $\gamma=0.1$, balanced	0.5451	0.32	0.33	0.34	0.6679
Neural Network (4-class)	(512,128,64), relu, sgd, lr=0.01, $\alpha=0.05$	0.5400	0.30	0.30	0.30	—
XGBoost (4-class+weights)	lr=0.1, n=200, depth=5, sub=0.8, col=1.0	0.6039	0.3525	0.3480	0.3598	—
AdaBoost (4-class)	DT(depth=3), lr=1.0, n=200, balanced	0.6194	0.3023	0.3081	0.3007	—
SVM (3-class)	rbf, C=1, $\gamma=0.1$, balanced	0.6081	0.41	0.42	0.40	0.6843
Neural Network (3-class)	(512,128,64), relu, sgd, lr=0.01, $\alpha=0.05$	0.5900	0.41	0.42	0.42	—
Random Forest (3-class)	n=300, depth=10, leaf=6, split=3, balanced	0.6099	0.47	0.46	0.48	—
XGBoost (3-class+weights)	lr=0.05, n=100, depth=7, sub=0.8, col=0.8	0.6004	0.4436	0.4389	0.4529	—
AdaBoost (3-class)	DT(depth=3), lr=1.0, n=200, balanced	0.6841	0.4077	0.4706	0.4068	0.72

6 Module 2: Advanced Regression

The Severity Impairment Index (SII) was modeled as a regression problem to preserve its ordinal

structure as a continuous target variable. The analysis used the outlier-cleaned dataset containing 8,417 samples and 31 features. An 80/20 train-test split was applied, and **Random Forest** and **XGBoost** regressors were evaluated. Hyperparameters were optimized using **GridSearchCV** and **RandomizedSearchCV**, with R^2 serving as the primary performance metric.

6.1 Random Forest Regressor

A two stage hyperparameter optimization strategy was applied to the Random Forest Regressor. **RandomizedSearchCV** was first used to efficiently explore a broad range of parameters, including the number of trees, maximum depth, minimum samples required for splitting and leaf nodes, and feature selection strategies. **GridSearchCV** was then employed to fine-tune the most promising parameter combinations. The final **4-class model** achieved an MAE of 0.52, RMSE of 0.676, and an R^2 score of 0.140 (**Figure 25**). The same procedure was also evaluated on a **3-class target** configuration, yielding similar performance (MAE = 0.51, RMSE = 0.652, R^2 = 0.146). As the improvement was marginal, the **4-class** configuration was **retained** as the final model.

The **relatively low R^2 values** indicate that only a limited proportion of the variability in SII can be explained by the available predictors. **Feature Importance** analysis identified Lifestyle_Risk_Score and SDS_Total_T as the strongest predictors, followed by physical and demographic variables, indicating that SII severity is influenced by a combination of behavioral, psychological, and physical factors. Summary metrics are reported in **Table 7**.

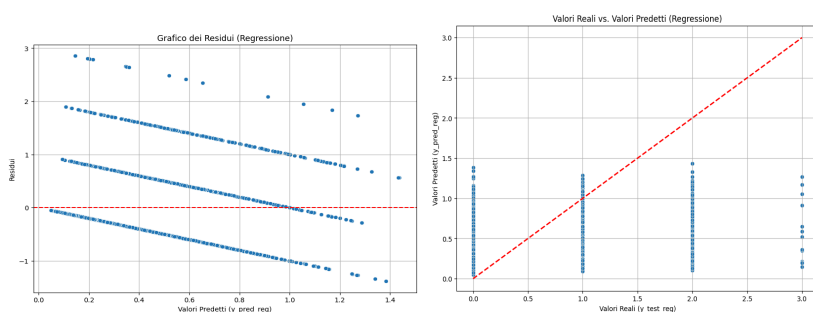


Figure 25: Residual Graph and Actual vs Predicted SII

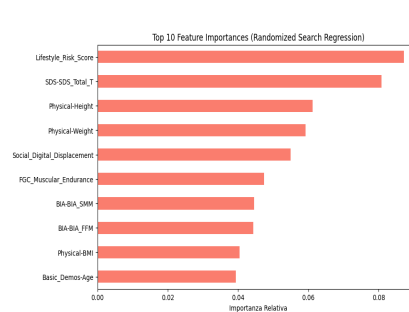


Figure 26: Feature importance

6.2 XGBoost Regressor

The XGBoost Regressor was trained to predict SII as a continuous target by minimizing RMSE. Hyperparameter tuning explored combinations of $n_estimators$ {100, 200}, max_depth {3, 5}, $learning_rate$ {0.05, 0.1}, and $subsample$ {0.8, 1.0}. The optimal configuration ($n_estimators=100$, $max_depth=3$, $learning_rate=0.05$, $subsample=1.0$) achieved a cross-validated R^2 of 0.1436. **Figure 25** shows a tendency toward lower SII predictions, reflecting the dominance of low-SII cases in the dataset. **Feature importance** analysis in **Figure 26** identified Lifestyle_Risk_Score as the strongest predictor, followed by Height, Age, and SDS_Total_T. LIME explanations further confirmed that lower distress levels and younger age contributed to lower predicted SII values, while higher lifestyle risk increased predicted impairment severity. These findings indicate that lifestyle and behavioral factors provide useful predictive information, although a substantial proportion of SII variability remains unexplained. Summary metrics are reported in **Table 7**.

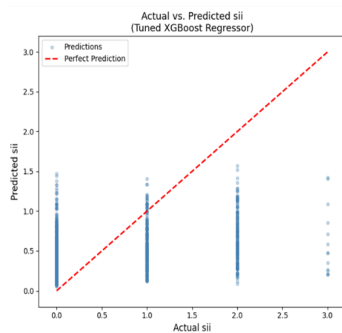


Figure 27: Residual Graph and Actual vs Predicted SII

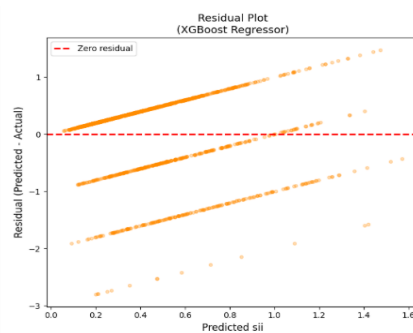


Figure 28: Feature importance

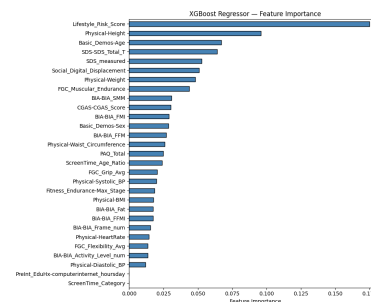


Table 6: Hyperparameter Grid and Performance

Model	Best Parameters	MAE	RMSE	R ²
XGBoost Regressor	n_estimators=100, max_depth=3, learning_rate=0.05, subsample=1.0	0.5250	0.6758	0.1420
Random Forest Regressor	n_estimators: 200, min_samples_split: 5, min_samples_leaf: 4, max_features: 'sqrt', max_depth: 12	0.521	0.676	0.142

The **low R² (0.1420)** reflects that physical and fitness data alone cannot reliably predict SII, capturing under 15% of the variance. This underscores the need for richer behavioral indicators, such as screen time, sleep patterns, and social media usage, to better explain internet use severity.

6.3 Advanced Regression for Physical BMI Prediction

BMI is an important health indicator associated with SII severity, as screen-based behaviour and sedentary lifestyles are often linked to increased BMI and poorer physical health. By regressing BMI from fitness, biometric, and behavioural features while excluding variables that could introduce target leakage, such as Physical-Weight, Physical-Height, it was evaluated whether the physical risk profile associated with problematic internet use can be predicted from objective measurements.

Random Forest and XGBoost

To maintain consistency with the SII regression task, an 80/20 train-test split (random_state = 42) was used for both Random Forest and XGBoost. Random Forest was optimized using a two-stage RandomizedSearchCV followed by GridSearchCV, while XGBoost was tuned using GridSearchCV. The two models achieved very similar performance, with R² values around 0.54, indicating that the available features provide a moderate but consistent signal for predicting BMI. Random Forest achieved an R² of 0.5353 (MAE = 1.7044, RMSE = 2.9005), while XGBoost achieved a slightly higher R² of 0.5393 (MAE = 1.6840, RMSE = 2.8881). Residual plots show errors centered around zero, although both models were less accurate for extreme obesity cases (BMI > 35), which are relatively rare in the dataset.

Feature importance analysis identified body composition measures as the strongest predictors, including fat mass, fat-free mass, skeletal muscle mass, age, and other anthropometric characteristics. These findings are physiologically meaningful because BMI is directly related to body composition. Several of the same variables were also among the most important features in the SII analysis, suggesting a shared relationship between body composition, lifestyle factors, and overall health status. **Figures 29–31** present the actual-versus-predicted plots, residual diagnostics, and feature importance rankings.

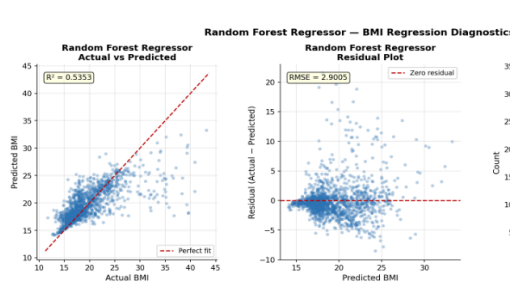


Fig 29: Random Forest Regressor-BMI diagnostics

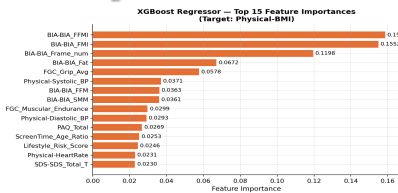


Fig 31: Feature importance comparison-Random Forest vs. XGBoost

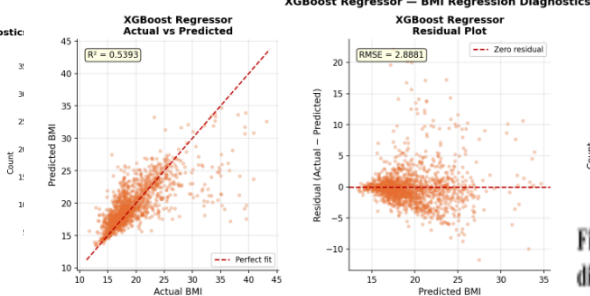


Fig 30: XGBoost Regressor- BMI diagnostics

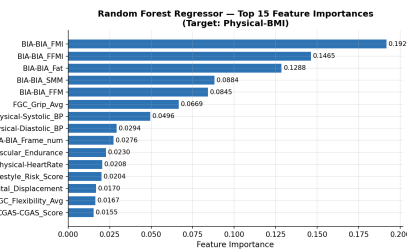


Table 7: Performance metrics for Physical-BMI prediction models

Model	MAE	RMSE	R²
Random Forest	1.7044	2.9005	0.54
XGBoost	1.6840	2.8881	0.5393

6.4 Connection to SII: Causal Chain and Residual Analysis

To further investigate the relationship between physical health and problematic internet use, Figure 37 shows that higher SII severity levels are associated with both increased BMI and greater daily internet use. BMI distributions shift progressively toward higher values across SII categories, indicating a positive relationship between body composition and problematic internet use.

BMI prediction error also increased with SII severity (None: 1.60, Mild: 1.87, Moderate: 1.98, Severe: 2.81 BMI units), suggesting that participants with more severe internet-use problems have more heterogeneous physical and behavioural profiles, making BMI harder to predict.

Under identical Random Forest settings, BMI regression ($R^2 \approx 0.54$) substantially outperformed SII regression ($R^2 \approx 0.14$), indicating that physical and fitness measurements contain a much stronger signal for body composition than for behavioural impairment. Overall, physical health factors are associated with SII but explain only a limited portion of its variability.

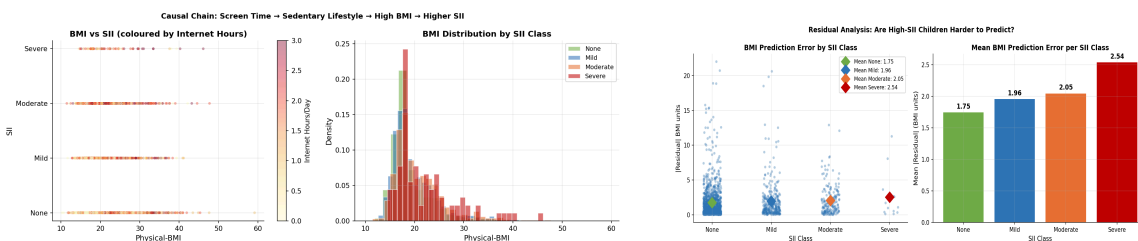


Figure 32. Relationship Between BMI, Internet Usage, and SII Severity.

7.Explainability

7.1 SHAP Analysis

To interpret the **Random Forest classifier**, TreeExplainer, a model-specific algorithm optimized for tree-based architectures, was applied to compute exact SHAP values for 200 representative test instances. Explanations were generated independently for each class to assess how specific features influence probability shifts across SII severity levels.

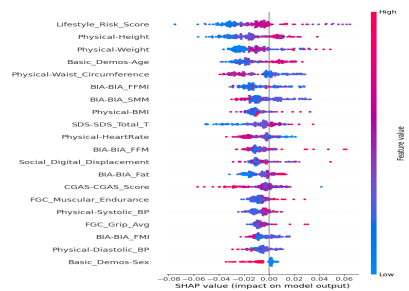


Fig 33: SHAP Analysis

Figure 33's beeswarm plot identifies **Lifestyle_Risk_Score** as the primary predictive driver. High values consistently push predictions toward the **higher SII** severity classes, while lower values favor the "None" (SII = 0) class, suggesting that problematic internet use is associated with a combination of screen-related behaviors, reduced physical activity, and emotional difficulties. Physical characteristics, including Height, Weight, Age, and Waist Circumference, are also among the most influential features, indicating that developmental and anthropometric differences contribute meaningfully to the prediction process.

In addition, detailed body composition measures derived from bioelectrical impedance analysis (FFMI, SMM, FFM, and Fat) provide information beyond simple body mass, allowing the model to distinguish differences in tissue composition and physical fitness profiles. **Psychosocial and health-related variables** further contribute to the predictions: higher SDS_Total_T scores (greater sleep disturbance) and lower CGAS_Score values (reduced psychosocial functioning) are consistently associated with increased predicted severity. Collectively, these patterns indicate that the model relies on a clinically coherent combination of behavioral, physical, and psychosocial factors rather than any single dominant variable.

Figure 34 complements this global interpretation through a **SHAP force plot**, illustrating how individual features either **increase (red)** or **decrease (blue)** the predicted probability for a specific instance. **Overall**, the SHAP analysis supports both the predictive validity of the Random Forest classifier and the feature engineering decisions made in Module 0 by demonstrating that the model's predictions are grounded in meaningful and interpretable feature relationships.

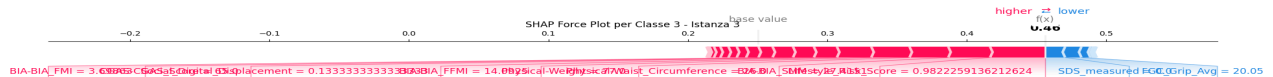


Figure 34: SHAP force plot showing feature contributions for Instance 3

7.2 LIME Analysis: XGBoost Regressor

To interpret the XGBoost Regressor's predictions at the local level, LIME was applied (Local Interpretable Model-agnostic Explanations), which explains individual predictions by fitting a simple linear model in the local neighbourhood of each instance. A LimeTabularExplainer was initialised on the training data in regression mode, and two contrasting instances were selected for explanation: a **low sii sample (actual sii=0.0)** and a **high sii sample (sii≥2)**, to understand how the model reasons differently across severity levels. For the **low sii instance (Figure 31)**, the model predicted 0.3753 against an actual value of 0.0. The dominant positive contributor was **SDS_Total_T > 56** (+0.1775), meaning elevated sleep disturbance pushed the prediction upward even for this low-severity child. **CGAS_Score** in the moderate range (61.5–65) and low

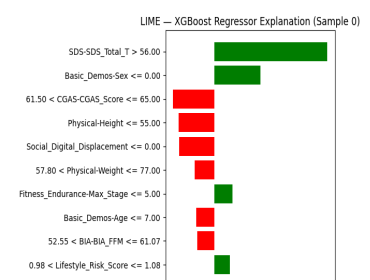


Fig 35: LIME for sii = 0

Social_Digital_Displacement both pulled the prediction downward, as did **young age (≤ 7)** and low physical measurements collectively keeping the prediction close to 0 and reflecting a generally healthy profile.

For the **high sii instance (Figure 36)**, the feature pattern shifts notably. **FGC_Muscular_Endurance > 8.5** was the strongest positive contributor, alongside **BIA_FFM > 71.08**, **older age (>12)**, **higher physical weight (77–104)**, and **BIA_SMM > 34.66**, all pushing the prediction toward higher severity. Interestingly, **SDS_Total_T** and **Lifestyle_Risk_Score** in moderate ranges pulled slightly downward for this instance, suggesting the model detected a mixed profile. Together, the two LIME explanations confirm that the XGBoost Regressor responds to clinically meaningful and instance-specific feature combinations rather than applying a uniform rule, complementing the global insights provided by the SHAP analysis on Random Forest.

7.3 LIME vs SHAP Comparison

7.3.1 Top features

Both methods identify behavioral and physical variables as key predictors of SII. **SHAP** globally highlights **Lifestyle_Risk_Score**, **Height**, **Weight**, **Age**, and **Waist_Circumference**. **LIME** provides a local view: in the examined low-SII instance, **SDS_Total_T** and **CGAS_Score** were among the most influential features, while the high-SII instance was primarily influenced by **FGC_Muscular_Endurance**, **BIA_FFM**, and **Age**. This confirms that while global feature importance remains stable, the model adapts its reasoning to **individual cases**.

7.3.2 Direction and Magnitude of Influence

Both methods consistently align: higher **Lifestyle_Risk_Score** and lower **CGAS_Score** are associated with increased SII severity, supporting the clinical plausibility of the model. Although SHAP and LIME use different explanation mechanisms and produce non-comparable contribution magnitudes, both converge on the same conclusion: SII predictions arise from a combination of behavioral, physical, and psychosocial factors rather than a single dominant variable.

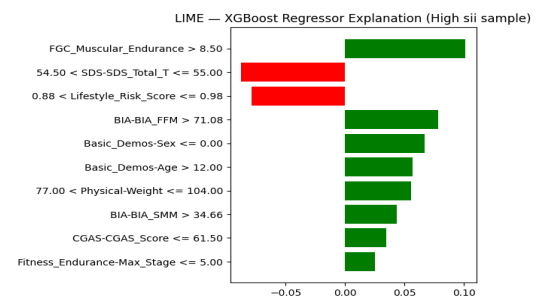


Fig 36: LIME XGBoost Regressor High SII Sample

8 Module 0: Data Understanding and Preparation for Time Series Dataset

8.1 Dataset overview and Preprocessing Pipeline

The time series dataset contains 4,437 recordings from 494 unique subjects, each comprising 200 timesteps sampled at 30-minute intervals (approximately 4.2 days), capturing physical activity through **enmo** and **anglez**. The binary target variable, **sii_binary** (0 = non-problematic, 1 = problematic), is moderately imbalanced: **66% non-problematic, 33.4% problematic**

Signal categories include:

- **behavioural signals** ['X', 'Y', 'Z', 'enmo', 'anglez', 'light', 'battery_voltage']
- **quality indicator** ['non_wear_flag'] + ['low_battery_flag']
- **metadata** ['id', 'sii_binary', 'weekday', 'quarter', 'relative_date_PCIAT']

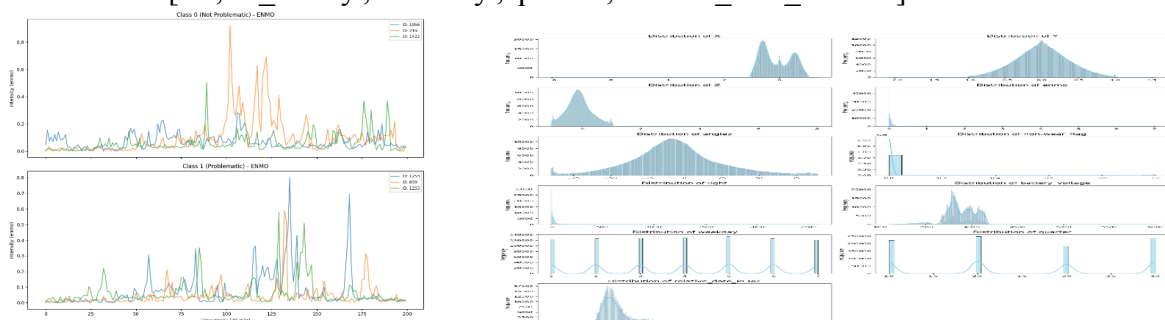


Fig 37: PAA of enmo Accelerometry data over 190 timesteps (5-hour segments)

8.2 Preprocessing Pipeline

Non-wear handling:

A total of **17 series** consisting entirely of non-wear periods (all 200 timesteps) and 112 series containing more than 50% non-wear observations were removed, reducing the dataset from 4,437 to 4,308 valid series. For the remaining recordings, physical signals (X, Y, Z, enmo, and anglez) were set to NaN at timesteps where **non_wear_flag > 0.5**, while light and battery-related metadata were retained.

Zero / Near-Zero Analysis

Near-zero ENMO values (< 0.02) were confirmed to represent genuine inactivity (such as sleep or stillness) rather than data corruption through comparison of inactivity fraction distributions across SII classes.

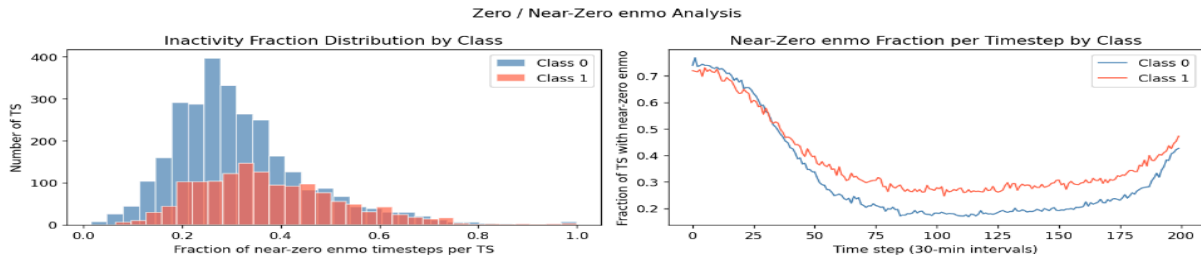


Fig 38: Distribution of inactivity by class

Gap-Limited Interpolation

Short NaN gaps (≤ 5 steps, **2.5 hours**) were filled using linear interpolation, while longer gaps were filled using forward/backward propagation to preserve behavioral continuity. A final validation confirmed that **no missing values** remained, and the quality flags (max_consec_gap, long_gap_filled) were updated accordingly.

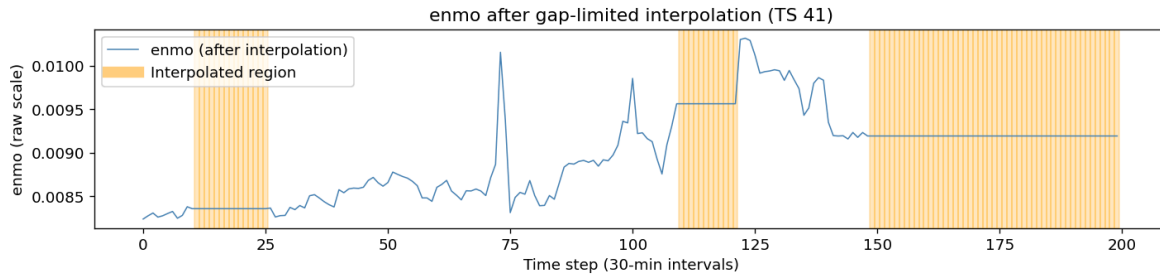


Fig 39: example of Gap-Limited Interpolation in enmo

Signal correlation and Subject-Level Train/Test Split

The correlation matrix revealed a strong correlation between Z and anglez, leading to the removal of Z to reduce redundancy. To prevent data leakage, a subject-level train/test split was performed using GroupShuffleSplit, ensuring that all recordings from a given subject were assigned exclusively to either the training or test set. An 80%/20% split was used.

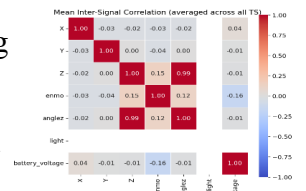


Fig 40: Mean Inter-Signal Correlation(average across all TS)

8.10 PAA Compression

Piecewise Aggregate Approximation (PAA) compressed each **200-step time series into 20 segments** (10 timesteps per segment, equivalent to 5 hours), reducing dimensionality from (N, 200, 7) to (N, 20, 7) while preserving the overall shape of the activity signals and reducing computational complexity.

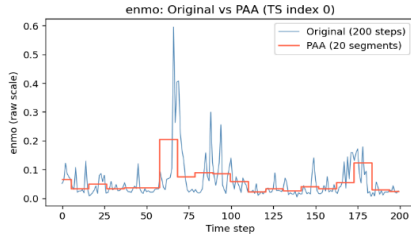


Fig 41: PAA compression

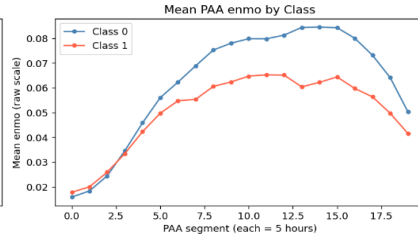


Fig 42: PAA reconstruction with 20 segments

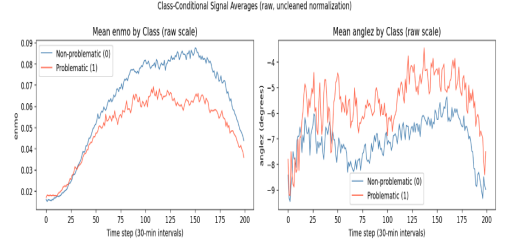


Fig 43: Class Conditional Signal Averages

8.11 Statistical Feature Extraction

A total of **45 statistical features** were extracted from each time series, including summary statistics for all 7 signal channels. Normalization was intentionally omitted at this stage to support task-specific preprocessing in later modules.

9 Module 3: Time Series Analysis (Motifs/Discords)

9.1 Introduction and Methodology

This section analyzes motifs and discords in the CMI actigraphy dataset using matrix profiles. A **total of 3,355 training series from 494 subjects (65.3% non-problematic, 34.7% problematic)** were analyzed to investigate whether problematic internet use is associated with distinct recurring behavioral patterns or anomalous activity episodes. The methodology includes class-conditional consensus motifs, regularity scores, Fisher exact tests for discord enrichment, and joint discord detection across the **ENMO** and **ANGLEZ** channels.

Analysis was performed on the cleaned Module 0 output. Each time series was independently **Z-score normalized per channel** before matrix profile computation, ensuring that similarity was determined by pattern shape rather than signal magnitude. Light and battery voltage channels were excluded because they represent environmental or device-related information rather than behavior.

Window Length Selection

A **representative subject** (Subject 2064, $sii = 1$, $enmo \sigma = 0.512$) was selected from the high-quality recordings for window-length optimization. The recording exhibits both pronounced activity bursts and quieter periods, providing sufficient behavioral variation for evaluating candidate window lengths. Window length m was optimized by maximizing the matrix profile separation score across $m \in \{10, \dots, 60\}$. The highest score was obtained at $m = 10$ (0.749), corresponding to a **5-hour window**. This window length provided the best separation between recurring patterns and anomalies and was subsequently applied to all 3,355 training series.

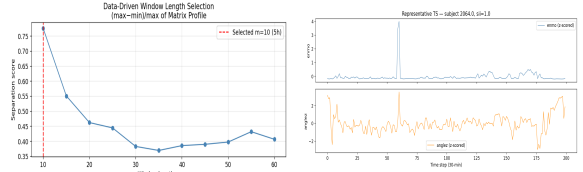


Fig 44: choose of window length selection

9.2 Motif and Discord Discovery

The top motif exhibited a near-zero distance, indicating near-perfect repetition of the same low-amplitude activity pattern which is consistent with sustained rest or sedentary periods recurring throughout the day, the most common behavioral state in wrist accelerometer data where the sensor is worn but the child is largely stationary. The second, third motifs also correspond to flat, near-zero activity periods at different points in the recording, confirming that low-activity periods are the main repeating pattern in the data. This is expected because, during the approximately 4.2 days of recording, children spend much of their time sleeping, resting, or being sedentary. Therefore, the most frequently repeated 5-hour windows naturally represent inactivity. Importantly, the same result was found even when analyzing the most

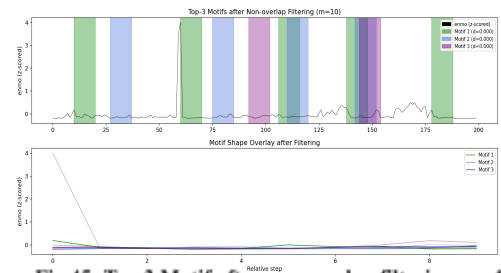


Fig 45: Top 3 Motif after non overlap filtering, m=10

active subjects, showing that sedentary periods remain the dominant recurring pattern regardless of how active a child is overall.

The **top discord** ($d=2.874$) represents a structurally isolated subsequence corresponding to an abrupt, high-amplitude enmo spike. The second ($d=2.562$) and third ($d=2.547$) discords mark additional regions of distinct, mild activity fluctuation. Notably, the visually prominent spike at timestep ~ 65 is not ranked top because multiple overlapping windows partially capture it, moderating its nearest-neighbor distance, illustrating that visual prominence and structural uniqueness are not the same.

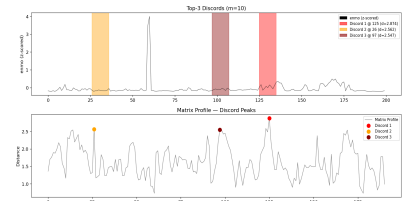


Fig 46: Top 3 Discord ($m=10$) and visualization of peaks

Ultimately, the **comparison between motifs and discords** highlights their different roles within the time series. Motifs represent recurring behavioral patterns that appear multiple times throughout the recording, whereas discords correspond to isolated subsequences that differ substantially from the rest of the series and have no close match elsewhere in the recording.

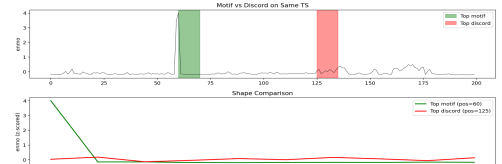


Fig 47: Motif vs Discord comparison

9.3 Global Analysis: Behavioral Regularity and Consensus Motifs by Class.

9.3.1 ENMO Regularity and Discord Analysis

Matrix profiles were computed for all 3,355 training series ($m = 10$), extracting regularity scores (minimum matrix profile value) and discord distances (maximum matrix profile value). Comparisons between Class 0 and Class 1 revealed no significant differences in regularity (0.668 vs. 0.655, $p = 0.057$) or discord distance (2.764 vs. 2.766, $p = 0.485$), indicating that movement intensity patterns are broadly similar across classes.

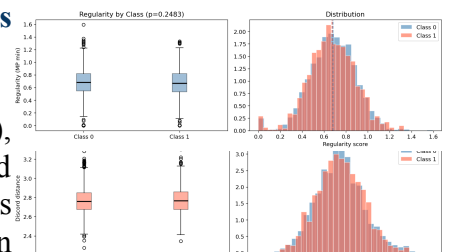


Fig 48: Boxplot of [variable] by group (0 vs 1), displaying five-number summaries (min, Q1, median, Q3, max)

9.4.2 ANGLEZ Regularity and Consensus Motifs

Both classes produced U-shaped consensus motifs with flat centers, reflecting the dominance of sedentary behavior. The L2 distance between class consensus motifs was 0.096 for ENMO and 0.202 for ANGLEZ. Although substantial overlap remained, the greater inter-class separation observed in ANGLEZ suggests that wrist orientation captures more class-related information than movement intensity at the 5-hour timescale.

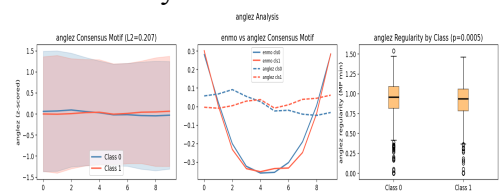


Fig 49: Anglez analysis

The most important result emerged from the regularity analysis. Non-problematic subjects exhibited significantly higher ANGLEZ regularity scores (0.938 vs. 0.913, $p = 0.0006$), indicating more consistent wrist orientation and stable postural routines over time. This was the only statistically significant class difference identified across the motif and discord analyses.

9.5 Discord-Based Class Analysis

9.5.1 Fisher Exact Test for Discord Enrichment

To test whether the most behaviorally anomalous recordings were disproportionately associated with problematic internet use, a Fisher exact test compared the top 10% highest-discord recordings ($n = 336$) against the remaining 90% ($n = 3,019$). Class 1 representation was virtually identical across both groups (33.0% vs. 34.8%), yielding $OR = 0.922$ and $p = 0.764$, indicating no significant enrichment. Children with the most structurally unusual activity patterns were no more likely to have problematic internet use than those without, suggesting that strong discords reflect normal individual variability rather than a characteristic associated with internet use severity.

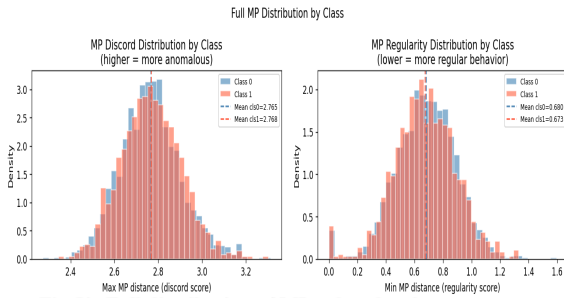


Fig 50: Full distribution of MP values by class

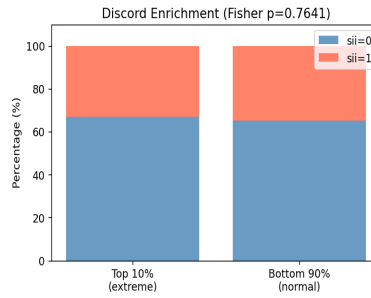


Fig 51: Discord Enrichment Test

Discord Density Over Time

Discords were most common in the early part of the recordings (0–66), accounting for approximately 39% of cases in both classes, and became less frequent later in the recording. Only minor differences (2–3%) were observed between the classes, indicating that the timing of anomalous activity patterns is not related to SII status.

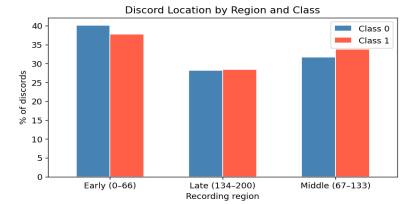


Fig 52: Discord Density over time

Joint enmo + anglez Discord Detection

Joint discords on timesteps where enmo and anglez simultaneously breach a 6-hour tolerance window were comparable across classes ($p=0.582$), confirming that co-occurring multichannel anomalies are not class-informative. Combined with the anglez regularity finding, this suggests that postural consistency in recurring patterns, not the occurrence of simultaneous activity and posture anomalies, is the more discriminative signal in this dataset.

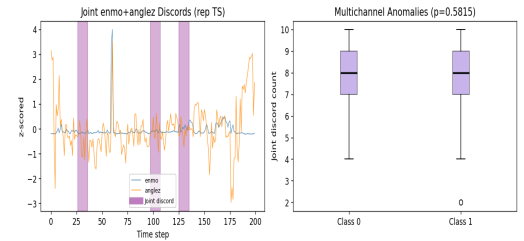


Fig 53: Timesteps where both enmo and anglez are simultaneously anomalous

Motif Generalization and Classification Utility

cross-Series Motif Matching

To test the generalizability of recurring patterns, the representative subject's top motif was searched against all training series using sliding-window Euclidean distance. The best cross-subject match ($\text{dist}=0.030$) occurred in TS 984 ($\text{sii}=0$). This validates that the pattern is behaviorally genuine rather than a recording artifact, confirming that shared activity archetypes exist across subjects independent of clinical class labels.

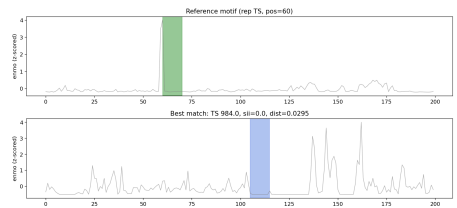


Fig 54: Cross Matching Motifs

Shapelet Bridge: Consensus Motif as Primitive Shapelet

Using the class-0 consensus motif as a shapelet template, a median-threshold classifier achieved accuracy of 0.521, marginally above chance with near-complete overlap between class distance distributions. This confirms that motifs capture recurring behavioral patterns but lack discriminative power.

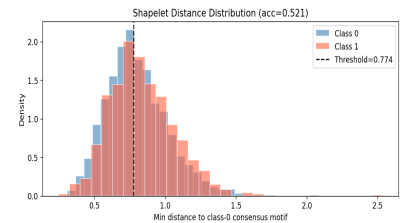


Fig 55: Shapelet Distance Distribution

Discussion

The principal finding identifies wrist posture regularity (**anglez**) as more class-discriminative than activity intensity (**enmo**) at the 5-hour scale; non-problematic subjects exhibit significantly more consistent postural routines ($p=0.0005$). Conversely, enmo regularity, discord enrichment ($p=0.764$), joint multichannel anomalies ($p=0.582$), and consensus motif classification ($\text{acc}=0.521$) all yielded null results. These findings demonstrate that behavioral anomalies are

common across classes rather than markers of internet use severity. Because problematic use lacks distinct 5-hour patterns, the classification module is motivated to adopt richer feature representations and learned shapelet approaches.

10. MODULE 2: CLUSTERING

10.1 Introduction and Preprocessing Paradigm

This module utilizes unsupervised learning to discover natural behavioral phenotypes from multi-channel wrist actigraphy data. The dataset includes 3,355 subjects (200 temporal steps, 7 channels), with a primary focus on ENMO (intensity) and ANGLEZ (postural orientation). The binary clinical target (sii) is moderately imbalanced (2,192 unimpaired vs. 1,163 impaired).

Preprocessing:

- **Scaling:** Global standard scaling is applied to each channel to preserve absolute intensity variations between waking and sleep states, preventing local normalization bias.
- **Outlier Control:** Winsorization (± 4 std dev) manages high-frequency artifacts.

Data Representations:

- **PAA-flattened vectors (40-dim):** Enables Euclidean K-Means clustering.
- **Raw ENMO sequences:** A class-stratified 1,500-subject subsample is used to compute Dynamic Time Warping (DTW) distances, accommodating temporal misalignments.

Statistical-behavioral features (45 variables): Captures signal distribution, entropy, and temporal organization.

Evaluation:

We evaluate **K-Means** and **Hierarchical Agglomerative Clustering** using Silhouette scores, Davies–Bouldin Index, and Adjusted Rand Index (ARI). Interpretability is assessed via centroid profiling, clinical cross-tabulation, and dimensionality reduction (PCA/t-SNE). Although clustering is performed without using class labels, the **SII labels** are later used to assess the **clinical relevance** of the discovered clusters

10.2 PAA + K-Means

Applied to the 40-dimensional PAA representation, K-Means clustering was evaluated for **k=2 to k=10** using the Silhouette score, **Davies–Bouldin index**, and **Adjusted Rand Index (ARI)**. Among the tested values, k=2 achieved the best clustering quality, producing the highest Silhouette score (0.2037) and the lowest Davies–Bouldin index (1.8606).

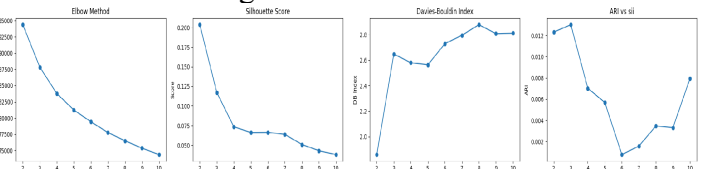


Fig 56: Clustering performance metrics across different k (Wiki Score, Silhouette Score, Davies-Bouldin Index, and ARI vs sii)

Silhouette values declined rapidly for larger k, reaching 0.0381 at k=10, indicating increasing cluster overlap.

The **final k=2** solution partitioned the dataset into **Cluster 0 (2,550 samples)** and **Cluster 1 (805 samples)**. The very low ARI (0.0123) indicates that the discovered clusters are largely independent of the SII labels. Therefore, the clustering captures natural behavioral variation rather than clinically meaningful impairment groups.

PCA and t-SNE visualizations further support this conclusion, showing substantial overlap between the clusters and no clear separating boundary. These results suggest that activity patterns vary along a continuum rather than forming distinct behavioral phenotypes associated with internet-use impairment.

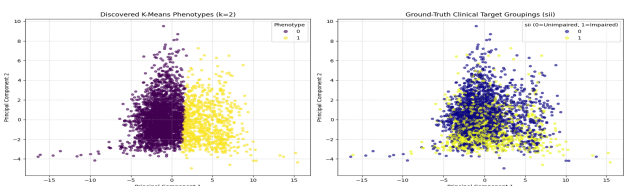


Fig 57: Cluster visualization by PCA and tSNE

10.3 K-Means on Statistical Feature

The 45-dimensional statistical-behavioral feature matrix summarizes each 200-step multichannel time series into a compact representation capturing distributional properties, signal complexity, and behavioral organization.

- **Temporal Central Moments:** Tracks kinetic volume and variance(Mean, Standard Deviation, Minimum, Maximum, Skewness, and Kurtosis)
- **Wave Complexity & Signal Energy:** Measures signal chaos and intensity (Entropy, IQR, RMS, Energy).
- **Behavioral Activity Proxies:** Captures activity patterns (Inactive/Active Ratios, Transition Counts, Circadian Strength).

This process produced a complete **3,355×45matrix** with no missing values. After **Z-standardization**, K-Means was evaluated across **k=2 to k=10** for cluster optimization.

The feature-space optimization loop confirms that **k=2** provides the most distinct natural cluster definitions. It captures a peak silhouette score of **0.2602** paired with a minimal Davies-Bouldin score of 1.3828. The final model separates the cohort into Cluster 0 (1,376 samples) and Cluster 1 (1,979 samples).

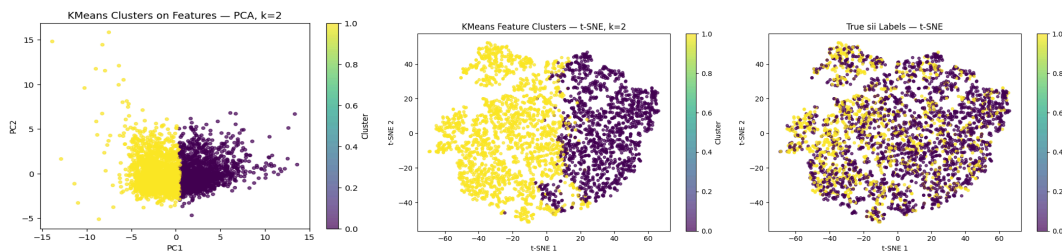


Fig 58: Cluster visualization by PCA and tSNE

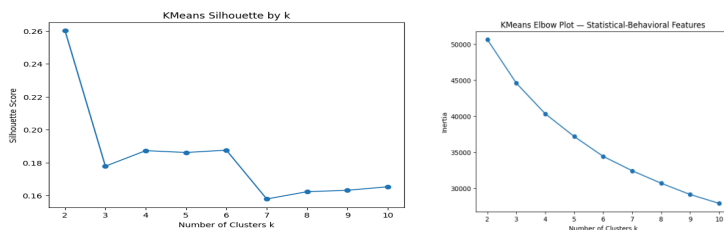


Fig 59-60: KMeans Elbow Plot and KMeans Silhouette by K

Cluster 0 (n=1,376): Predominantly unimpaired (77.5%). Characterized by high physical activity, movement variability, and strong circadian rhythms.

Cluster 1 (n=1,979): Higher impairment rate (43.2%). Characterized by lower activity and weaker circadian rhythms, but elevated posture dispersion (anglez_iqr) and extreme activity episodes (enmo_kurtosis).

Overall, **Cluster 0** represents a more active and regular behavioral profile, whereas **Cluster 1** is associated with lower activity levels and a higher prevalence of impairment.

Table 8. Comparison of clustering performance between PAA + Kmeans and Statistical feature methods.

Method	k	Silhouette	Cluster Size
PAA + Kmeans	2	0.2037	Cluster 0 (n=2,550) Cluster 1 (n=805 Samples)

Statistical feature	2	0.2602	Cluster 0 (n=1,376) Cluster 1 (n=1,979)
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10.4 DTW Hierarchical Clustering

DTW distances (Sakoe-Chiba band, window=10%) were computed on a 1,500-subject stratified ENMO subsample to capture temporal variability Euclidean distance cannot. Agglomerative hierarchical clustering was tested across k=2–10 with average, complete, and single linkage (Ward excluded invalid for non-Euclidean DTW matrices).

Table 9 DTW Hierarchical Clustering: Linkage Comparison (k = 2 and k = 3)

Linkage	k	Silhouette	ARI	Min Cluster Size
Average	2	0.3689	−0.0123	30
Average	3	0.3313	−0.0126	14
Complete	2	0.2473	0.0092	615
Complete	3	0.2407	0.0050	12
Single	2	0.4799	−0.0006	1
Single	3	0.3449	−0.0012	1

- **Single Linkage:** Highest Silhouette score (0.4799) but failed to capture meaningful patterns, isolating individual outliers while trapping the remaining samples in a single 'mega-cluster'.
- **Complete Linkage:** Produced balanced cluster sizes but resulted in weak boundary definitions (Silhouette = 0.2473).
- **Average Linkage:** The superior configuration, balancing a robust Silhouette score (0.3689) with stable, meaningful cluster sizes.

10.4.1 Final Model: Average Linkage, k = 2

The selected model (DTW hierarchical clustering, average linkage, k=2) partitions the 1,500-sample subsample into two distinct groups that reflect physical behavioral similarities rather than clinical impairment status (ARI = −0.0123):

- **Cluster 0 (Standard cohort):** 1,470 samples with a distribution (64.8% unimpaired, 35.2% impaired) closely mirroring the baseline dataset.
- **Cluster 1 (Specialized phenotype):** A small subgroup of 30 samples exhibiting a distinct temporal activity profile relative to the rest of the cohort, showing an impairment rate of only 10% (27 unimpaired, 3 impaired).
- The model's high-level bifurcation is visible in the truncated dendrogram (level 5), while t-SNE projections reveal clear, non-linear behavioral substructure.

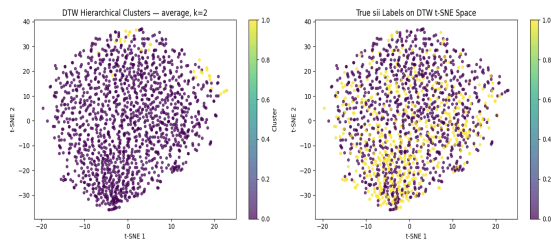


Fig 61: Hierarchical cluster visualization by PCA and tSNE

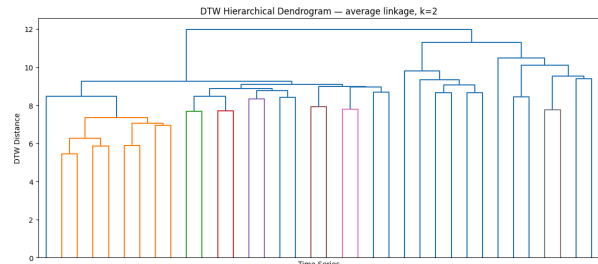


Fig 62: Dendrogram

10.5 Hierarchical Clustering on Statistical Features

A parallel agglomerative hierarchical clustering was evaluated over the standardised 45 feature matrix using all four linkage methods (Ward is valid here as feature vectors are Euclidean-aligned). Four linkage methods were compared: **Ward linkage**, **Complete linkage**, **Average linkage**, and **Single linkage**.

Table 10: Linkage Optimization for Agglomerative Feature Clustering ($k = 2$)

Linkage	k	Silhouette	ARI	Min Cluster Size
Ward	2	0.1893	0.0620	1366
Complete	2	0.7891	-0.0000	3
Average	2	0.7891	-0.0000	3
Single	2	0.8058	0.0005	1

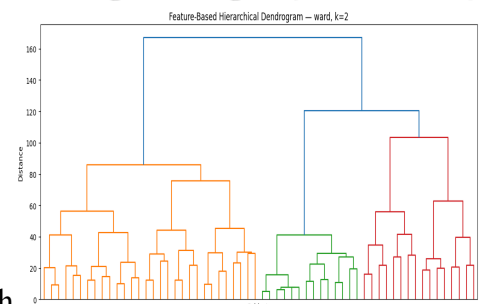
Single, Complete, and Average linkage methods all yielded inflated silhouette scores (ranging from 0.7891 to 0.8058), yet they proved statistically unstable and difficult to interpret due to severe chaining effects or excessive outlier sensitivity, often isolating outliers while collapsing the rest of the population into a single dominant group.

Ward linkage provided the most balanced and interpretable clustering solution:

- **Mechanism:** Minimizes within-cluster variance and effectively avoids chaining artefacts.
- **Configuration:** Optimal at $k = 2$ (Silhouette = 0.1893, DB = 1.7140).
- **Results:** Produced stable, balanced groups (1,366 and 1,989 subjects) and achieved the highest ARI (0.0620) of any method tested.

The methods are visualized using PCA and t-SNE. For each method, two panels are compared side by side: cluster assignment and true sii label. As shown on the figures below:

Fig 63: Dendrogram (statistical feature)



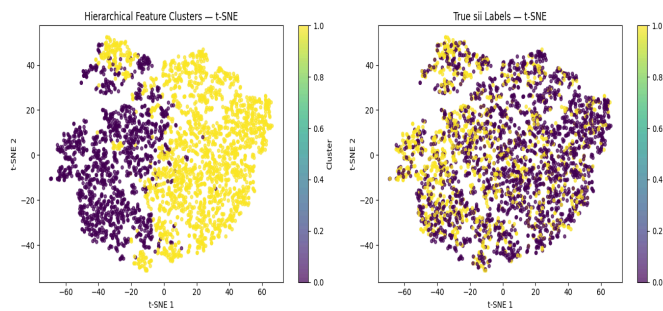


Fig 64: Cluster visualization by tSNE

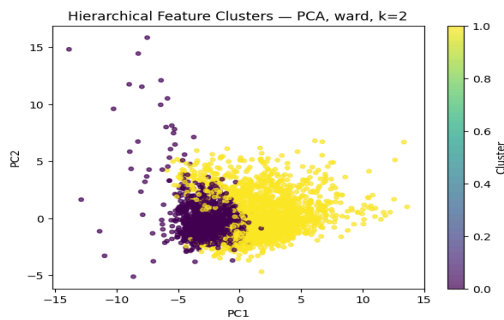


Fig 65: Cluster visualization by PCA

Across all methods, the cluster coloring never aligns perfectly with the sii coloring. This is not a methodological failure, it is a substantive finding confirming that wrist actigraphy alone does not cleanly partition subjects by internet-use severity.

10.5.1 Clinical Cross Tabulation

Table 11: Ward Linkage Cluster Cross-Tabulation Against Clinical Metadata

Cluster	Unimpaired (sii=0)	Impaired (sii=1)	Impairment Rate
Cluster 0: Sedentary / Rest	726	640	46.9%
Cluster 1: Volatile / Active	1,466	523	26.3%

The Sedentary/Rest cluster exhibits a substantially higher impairment rate (46.9%) than the Volatile/Active cluster (26.3%), suggesting a potential association between sedentary behavioral patterns and elevated clinical risk.

10.6 Comparative Summary and Conclusions

Table 12 Summary of All Clustering Methods

Experiment	Method	Best k	Silhouette	DB Index	ARI
PAA K-Means	K-Means (Euclidean)	2	0.2037	1.8606	0.0123
DTW Hierarchical	Agglomerative (Average)	2	0.3689	N/A	-0.0123
Feature K-Means	K-Means (Euclidean)	2	0.2602	1.3828	0.0178
Feature Hierarchical	Agglomerative (Ward)	2	0.1893	1.7140	0.0620

We have Four key conclusions:

- **Optimal Profiling:** Across all clustering experiments, a two-cluster solution provided the most interpretable and stable partition of the data, uniformly separating "active/dynamic" from "sedentary/rest" behavioral profiles.
- **Behavioral vs. Diagnostic:** Clustering recovers behavioral archetypes rather than clinical diagnostic categories (indicated by ARI values near zero); behavioral patterns are driven by age, health, and lifestyle rather than the SII target.

- **Methodological Rigor:** Algorithm selection significantly impacts results; Average and Ward linkage methods consistently outperform Single and Complete linkage, provided metrics are evaluated in the correct feature space.
- **Clinical Relevance:** Although clustering does not directly recover SII categories, the sedentary/rest phenotype consistently shows higher impairment prevalence (46.9% vs 26.3%), suggesting that actigraphy contains clinically relevant behavioral **information**.

11 Module 3: Time Series Classification

11.1 KNN

KNN classifiers were evaluated on normalised time series using two main approaches: **Euclidean distance** and **Dynamic Time Warping (DTW)**, each applied to both raw and PAA-transformed data.

11.1.1 KNN Classification with Euclidean Distance (Normalized and PAA)

KNN classification with Euclidean distance yielded similar results across both datasets, with persistent challenges in minority class detection:

- **Raw Time Series:** With StandardScaler and K=5 (uniform weights), the model achieved 66.6% accuracy and a 0.54 Macro-F1.
- **PAA Dataset:** Normalization and K=7 (uniform weights) resulted in a slightly higher accuracy (67.2%) but a lower Macro-F1 (0.518).

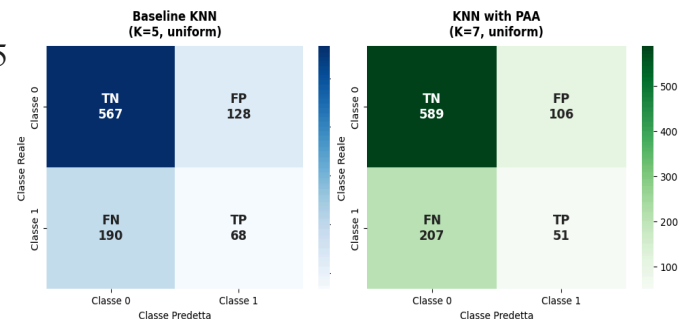


Fig 66: Confusion Matrix KNN (base)

Fig 67: Confusion Matrix KNN (PAA)

In both configurations, performance on the minority class remained limited, showing low precision and recall.

Table 13: Performance of KNN classification (Euclidean)

Method Euclidean	Acc.	Macro Prec.	Macro Rec.	Macro F1	W. Prec.	W. Rec.	W. F1	ROC-AUC
KNN Baseline	0.67	0.55	0.54	0.54	0.64	0.67	0.65	0.55
KNN + PAA	0.67	0.53	0.52	0.52	0.63	0.67	0.64	

11.1.2 KNN Classification with DTW Distance (Normalized and PAA)

KNN with DTW distance and Sakoe-Chiba constraints produced slightly stronger performance than Euclidean distance, particularly after PAA transformation.

- **Raw Time Series:** Using K=5 and a radius of 3, the model achieved 66% accuracy (Macro-F1 0.60, ROC-AUC 0.60).¹
- **PAA-Transformed:** Using K=5 and a radius of 5, the model improved to 71% accuracy (Macro-F1 0.60, ROC-AUC 0.62).¹

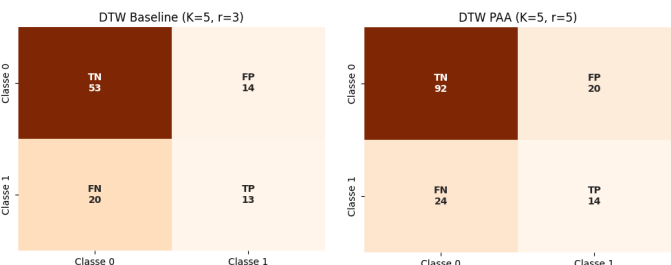


Fig 68: Confusion Matrix KNN (base)

Fig 69: Confusion Matrix KNN (PAA)

Table 14: Performance of KNN classification DTW

Method	Accuracy	Macro Prec.	Macro Rec.	Macro F1	Weighted Prec.	Weighted Rec.	Weighted F1	ROC-AUC
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DTW Baseline	0.66	0.60	0.59	0.60	0.65	0.66	0.65	0.60
DTW +PAA	0.71	0.60	0.59	0.60	0.70	0.71	0.70	0.62

Overall Conclusions:

KNN with Euclidean distance produced similar performance on both raw and PAA-transformed data, achieving approximately 67% accuracy but showing limited discrimination of the minority class. DTW-based KNN produced comparable or slightly stronger results, with the **DTW+PAA** configuration achieving the best overall performance (**Accuracy = 0.71, Macro-F1 = 0.60, ROC-AUC = 0.62**). These results suggest that combining temporal alignment through DTW with dimensionality reduction through PAA provides a more effective representation of behavioral time series. Nevertheless, minority-class detection remained challenging across all KNN configurations, indicating that additional feature learning or more expressive classification models may be required.

11.2 Shapelet-Based Classification and Motif Analysis

Shapelet Extraction was performed using the Random Shapelet Transform (RST) after reducing the input space from seven to three channels (0, 3, and 5) via Mann–Whitney U test. Twenty informative shapelets were selected and used for classification. Classification was performed using five shapelet-based models: **RST + Decision Tree, RST + Logistic Regression, RST + XGBoost, ShapeletTransformClassifier, and RDSTClassifier (aeon)**. Their performances are summarized in **Table 15**.

Performance was moderate across all models (Macro F1: 0.48–0.51), with the RDSTClassifier achieving the best results (71% accuracy, Macro F1 = 0.51, ROC-AUC = 0.53). Overall, shapelet-based features captured some temporal structure but provided only limited class separation.

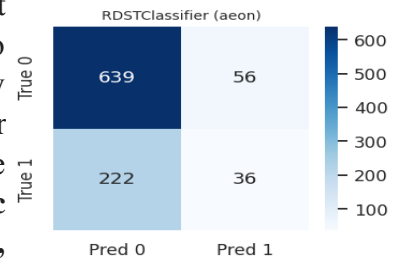


Fig 70(RDSTClassifier Confusion Matrix)

Table (15): Shapelet-Based Classification Performance Summary

Model	Accuracy	Macro F1	Roc-Auc
RST + DecisionTree	0.51	0.48	0.49
RST + Logistic Regression	0.54	0.50	0.55
RST + XGBoost	0.58	0.50	0.51
ShapeletTransformClassifier	0.69	0.49	0.53
RDSTClassifier (aeon)	0.71	0.51	0.53

11.3 Shapelet Interpretability & Alignment with Motifs/Discords

The **top 4 shapelets** by information gain all originate from channel 3 (ENMO), the dominant activity channel. They are relatively long (73–98 timesteps out of 200) and cluster in the early-to-mid portion of the recording (start positions 6–35). All four shapelets are associated with Class 0, indicating that the most informative discriminative subsequences correspond to low-impairment subjects. As shown in Table 16 and Fig. 71, Class 1 series tend to be more distant from these patterns, although the substantial overlap in distance distributions indicates only modest discriminative power. The decision tree analysis further suggests that classification depends on combinations of multiple shapelet-distance thresholds rather than a single dominant subsequence.

Table (16): Shapelets by Information Gain

Rank	IG	Length	Channel	Start Pos	Associated Class
1	0.01357	73	3	30	0
2	0.01334	93	3	14	0
3	0.01312	98	3	6	0
4	0.01205	94	3	35	0

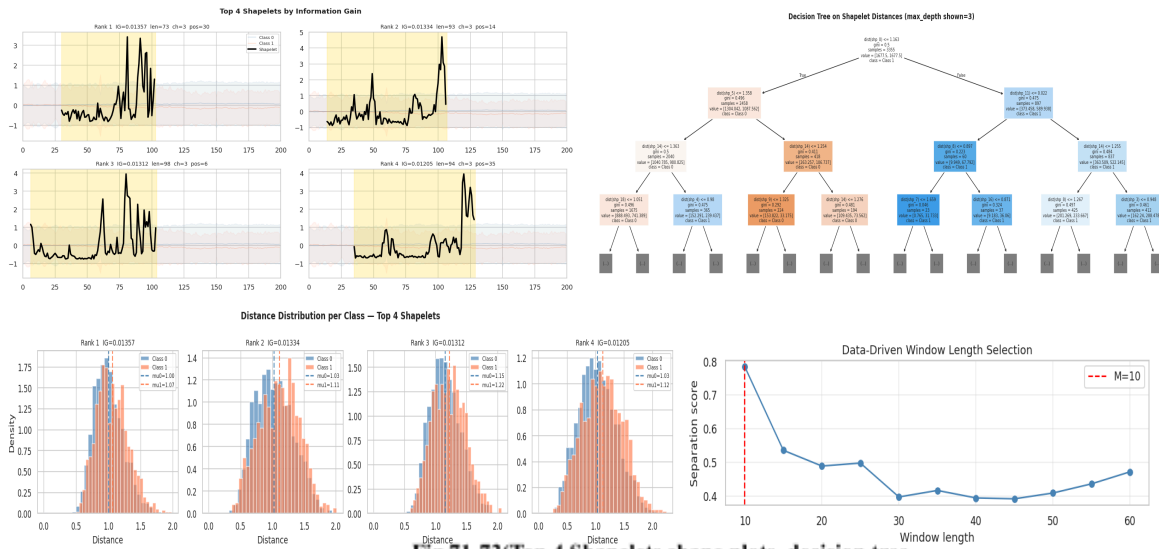


Fig 71-73(Top 4 Shapelets shape plots, decision tree, distance distributions & window length selection)

Table17(Motif Occurrences)

Motif	Occurrences	Best MP Distance	Positions
1	4	0.0000	[9, 45, 137, 177]
2	6	0.0000	[1, 37, 70, 111, 143, 190]
3	5	0.0000	[24, 60, 106, 138, 178]

Across all 3,355 training instances, ENMO regularity and discord distance showed no significant differences between SII classes. However, anglez regularity exhibited significant class separation, suggesting that postural consistency may be more informative than movement intensity.

11.4 Relationship Between Shapelets and Motifs

Matrix profile analysis using **STUMPY** on the **ENMO** channel identified recurring patterns and anomalies, with a data-driven window length selection of $m = 10$. Three motif families and three discords were discovered, with the motifs exhibiting near-exact self-matches that indicate highly repetitive behavioural patterns. The most frequent motif recurred at approximately 30-step intervals, suggesting an underlying periodic activity rhythm.

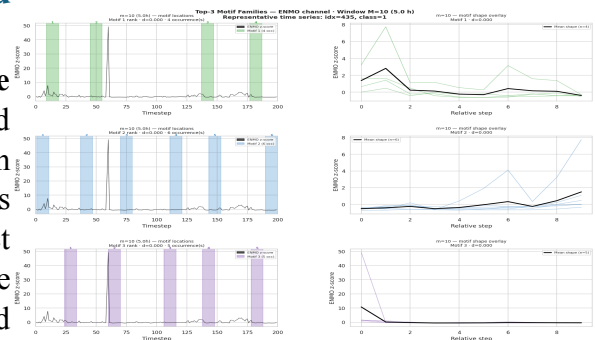


Fig 74(Top 3 Motif Families vs. Shapelet alignment, ENMO channel)

To evaluate their discriminative value, the **top RST shapelet (length 73, start = 30)** was compared with the discovered motifs. It spatially overlapped only with Motif 3, while the remaining motifs fell outside the shapelet window, indicating that the motif captured only a fragment of the discriminative pattern. Using the Class-0 consensus motif as an unsupervised shapelet yielded only weak classification performance (**Accuracy = 0.54, Macro F1 = 0.53**), confirming that recurrence alone is insufficient for class discrimination and that supervised shapelet extraction provides additional predictive information beyond unsupervised motif discovery. **Discords** similarly showed no meaningful class separation. Across all 3,355 training instances, ENMO motif regularity and discord distance did **not differ significantly** between SII classes. **In contrast**, anglez regularity exhibited significant class separation, suggesting that postural consistency may be more informative for impairment prediction than movement intensity alone.

11.5 Kernel-Based Classifier: MiniROCKET

MiniROCKET pairs a fixed convolutional transform with a RidgeClassifier, offering strong benchmark performance at low computational cost. Z-score normalization was applied independently per series per channel to remove amplitude variation, directing the model toward structural patterns.

Four configurations were tested on the 2-channel input (enmo, anglez):

- **V1 (Shape):** Raw Z-scored data which captures geometric trajectories and spatial variation
- **V2 (Velocity):** First-difference post-standardization ($\Delta X_t = X_t - X_{t-1}$) that isolates acceleration and abrupt behavioral spikes
- **V3 (Moving Average):** Rolling window ($w=6$) which captures slow trends and smoothed cycles
- **V4 (Multi-View):** Concatenation of V1 and V2 feature matrices that allows the linear model to jointly optimize over shape and velocity spaces simultaneously

Each configuration used a MiniROCKET along RidgeClassifier pipeline with $\alpha=100.0$, selected via 5-fold stratified cross-validation on the training split, with balanced class weighting to address the 2:1 class imbalance. All configurations were evaluated on the held-out test set (N=953).

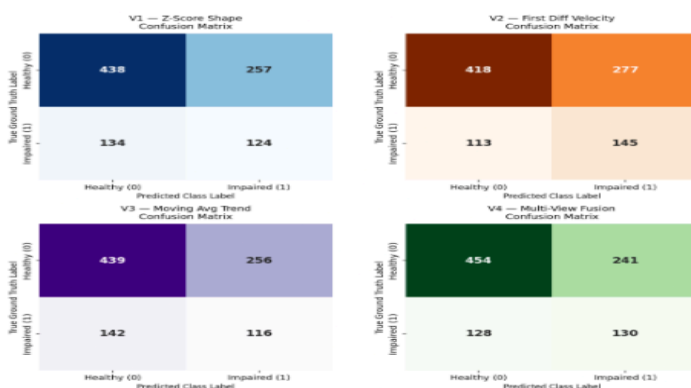


Fig 75(MiniROCKET Confusion Matrices V1-V4)

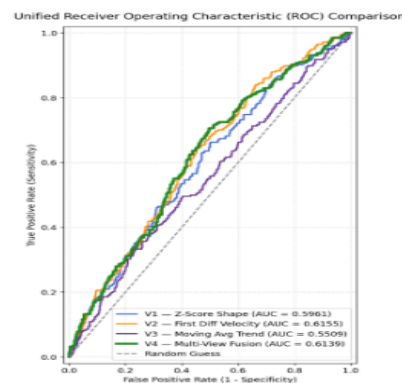


Fig 76 (Unified Receiver Operating Characteristic (ROC) Comparison)

Table18 MiniROCKET: Performance by variant

Variant	Accuracy	Macro Prec.	Macro Rec.	Macro F1	Weighted F1	ROC-AUC
V1: Z-score (Shape)	0.5876	0.5524	0.5615	0.5416	0.6082	0.5854

Variant	Accuracy	Macro Prec.	Macro Rec.	Macro F1	Weighted F1	ROC-AUC
V2: First Diff (Velocity)	0.5761	0.5482	0.5601	0.5351	0.5984	0.5943
V3: Moving Avg	0.5834	0.5421	0.5459	0.5312	0.6030	0.5528
Multi-View Fusion (V4)	0.6128	0.5658	0.5791	0.5622	0.6312	0.6015

Across all **four configurations**, no single transformation dominates in isolation ,each captures a different aspect of the actigraphy signal. **Raw shape (V1, 0.5876)** establishes that geometric structure carries discriminative information, but is insufficient alone. **Isolating velocity (V2, 0.5761)** marginally hurts performance, indicating that abrupt behavioral spikes are informative but noisy when stripped of shape context. **Heavy smoothing (V3, 0.5834)** recovers slightly but discards local volatility that the model benefits from. The **multi-view fusion (V4, 0.6128)** achieves the best result precisely because **neither shape nor velocity alone** fully characterizes impairment-related behavior ,their combination allows the ridge classifier to exploit complementary temporal qualities, yielding a more generalizable decision boundary.

Conclusion

The classification pipeline evaluated several advanced configurations to predict internet use severity. **KNN with DTW+PAA** is the top performer with **0.71 accuracy and Macro F1 of 0.60**. The reason DTW outperformed all other approaches is fundamentally about the nature of actigraphy data itself is that children's activity patterns are temporally irregular, meaning the same behavioral routine (a rest cycle, an activity burst) can occur at slightly different times across recordings, and DTW's elastic matching absorbs these phase shifts while **Euclidean distance** penalizes them harshly, explaining the consistent gap between the two distance metrics. Shapelet methods despite their theoretical appeal failed to find class-discriminative subsequences not because the method is weak but because problematic internet use simply does not leave a consistent short-window fingerprint in wrist movement ,the behavior is diffuse and distributed across the entire recording rather than concentrated in any localized pattern. **MiniROCKET's multi-view fusion** being the best among fixed-transform classifiers tells a similar story that the model only improved when it stopped looking at shape or velocity in isolation and started reasoning about both simultaneously, which reflects that **impairment related behavior** is a complex temporal texture, not a single signal characteristic.

11.5.1 Time Series Conclusion

The entire time series project shows a clear mismatch between the type of data collected and the clinical question being asked. **Wrist actigraphy** measures how a child moves, but **internet addiction** is influenced by cognitive involvement, social reliance, and behavioral compulsion. A child addicted to screens does not necessarily move differently than a healthy child; they might sit still while studying or while scrolling through their device, and accelerometry cannot tell the difference. This is not a flaw in the method but a limitation of the structure, and every analytical layer in this project confirmed this independently. The most significant evidence came not from the classifiers but from the **unsupervised analysis**. **Clustering** across four different methods consistently uncovered activity patterns instead of clinical categories. The sedentary group's higher impairment rate of 46.9% compared to 26.3% in the active group indicates a probabilistic risk connection, not a strict diagnostic line. The one truly distinguishing signal, **angle postural regularity**, suggests something clinically relevant: **non-problematic children** tend to have more structured and consistent routines, hinting that circadian health and lifestyle regularity are the protective factors rather than just higher physical activity levels. This subtle but crucial distinction

should be explored more directly in future research. We should build features that specifically capture routine consistency and temporal structure instead of merely raw movement statistics.

Ultimately, these should be combined with the behavioral and psychosocial features, where the real predictive signal for SII has already been shown to exist.